

# Robust variance estimation for multiple imputation using predictive mean matching or parametric imputation models

Suyang Yu<sup>1,\*</sup>, Lucy D’Agostino McGowan<sup>2</sup>, and Brian D. Williamson<sup>3,4,5</sup>

<sup>1</sup>Department of Statistics, University of Washington

<sup>2</sup>Department of Statistical Sciences, Wake Forest University

<sup>3</sup>Biostatistics Division, Kaiser Permanente Washington Health Research Institute

<sup>4</sup>Department of Biostatistics, University of Washington

<sup>5</sup>Vaccine and Infectious Disease Division, Fred Hutchinson Cancer Center

\*Corresponding Author. Email: [yu000434@uw.edu](mailto:yu000434@uw.edu)

September 10, 2026

## Abstract

Multiple imputation (MI) is widely used for handling missing data in medical research. After creating multiple completed datasets, the standard procedure averages the completed-data point estimates and estimates their variance using Rubin’s rules (RR). Although RR is attractive because of its simplicity, its validity depends on congeniality between the imputation and analysis procedures. When congeniality fails, the RR variance can be either too small or too large, leading to miscalibrated confidence intervals. Robins and Wang (RW) developed an estimating-equation variance estimator that remains valid without congeniality. Its practical use, however, has been limited by the lack of accessible software and by the absence of extensions to commonly used chained-equation and predictive mean matching (PMM) workflows. We address these barriers by developing the open-source R package `rw`, extending the RW variance estimator to chained conditional imputation models fitted through `mice`, and proposing an RW variance procedure for PMM. The PMM extension retains the ordinary hard-donor imputations produced by MICE: each realized donor is evaluated under a differentiable working donor law defined over all observed donors, and the analysis-score covariance accounts for dependence induced when multiple recipients copy values from the same observed donor. We evaluate the proposed methods in benchmark simulations based on the original RW setting and in validation-study simulations motivated by electronic health record data. Across scenarios in which RR was anti-conservative (by up to 23 coverage percentage points) or conservative (overstating standard errors by 30–50%), the proposed estimators produced confidence-interval coverage close to the nominal level, for both parametric and PMM imputation, without requiring analysts to change their usual MICE workflow.

# 1 Introduction

Multiple imputation (MI) is a standard tool in medical research for handling missing data in electronic health records, cohort studies, and clinical trials (Rubin, 1987). After creating several completed datasets, the standard approach is to average the completed-data point estimates and to estimate the variance using Rubin’s rules (RR), which add the within-imputation variance to a scaled between-imputation variance (Rubin, 1987; van Buuren and Groothuis-Oudshoorn, 2011; Sterne et al., 2009; White et al., 2011). The appeal of RR is practical: only the completed-data point estimates and their model-based standard errors are needed, and no special access to the imputation mechanism is required.

The validity of RR rests on a condition known as congeniality between the imputation and analysis procedures (Meng, 1994; Xie and Meng, 2017; Bartlett and Hughes, 2020): roughly, the two procedures must be embeddable in a single joint model, so that the imputer and the analyst make the same assumptions and use the same information. In a seminal paper, Wang and Robins (1998) established the asymptotic theory for a variance estimator based on RR: when congeniality holds, RR is asymptotically correct; when it fails, the RR variance can be systematically too small or too large. Congeniality can fail in either direction. Anti-conservative estimates arise when the analysis exploits structure that the imputation model does not reflect. Conservative estimates arise when the imputation model uses information or assumptions beyond those of the analysis—for example, auxiliary variables, or a model fitted to all subjects while the analysis is evaluated only in a subgroup—so that imputation-induced variation inflates the between-imputation component (Meng, 1994). Both patterns lead to confidence intervals that fail to maintain nominal coverage.

Two settings where congeniality fails are especially relevant in medical data analysis. The first is a structural mismatch between the imputation and analysis models: for instance, imputing under homoskedasticity when the analysis estimating equation allows heteroskedastic errors, or restricting the analysis to a covariate-defined subgroup when the imputation model was fitted on all subjects (Bartlett et al., 2015). The second is a threshold-by-imputation setting: in electronic health record (EHR) studies with error-prone measurements, the same imputed variable both enters the analysis and determines who is included in the analysis set. Here the imputation model is fitted on all subjects, while the analysis estimating equation is evaluated in a post-imputation subset that depends on the imputed value. Giganti and Shepherd (2020), hereafter GS, demonstrated this in an HIV cohort study where validated CD4 count simultaneously determined eligibility and served as an analysis covariate: RR overstated the variance by 29% relative to the correctly calibrated estimator, producing unnecessarily wide confidence intervals. The same issue arises whenever imputed variables determine study eligibility, which has become increasingly common as analyses rely on imputed inclusion criteria derived from imperfect EHR data; related EHR validation-and-imputation analyses raise the same calibration question (Giganti et al., 2020; Shen et al., 2026).

To avoid the requirement of congeniality, Robins and Wang (2000), hereafter Robins–Wang or RW, proposed an estimating-equation alternative to RR for estimating the MI variance. Rather than approximating the variance through RR, the RW estimator is based on three sources of variance: the sampling variability of the complete-data analysis score, the variability propagated from estimating the imputation model parameter, and the cross-covariance between the two. Because this construction targets the repeated-sampling variance of the implemented estimator, it remains consistent when a parametric imputation model is used, whether or not the

imputation and analysis procedures are congenial. Its guarantee concerns variance calibration alone: when the imputation model is misspecified, the point estimator may be biased for the scientific target, and this bias is unaffected by the choice of variance estimator. We formalize this distinction in Section 2.2. An alternative route to valid inference under uncongeniality is the bootstrap, which re-runs the entire imputation and analysis procedure across resamples (Schomaker and Heumann, 2018; Bartlett and Hughes, 2020); we do not pursue it here because repeating the full imputation workflow can be computationally prohibitive in large studies.

Despite these advantages, the RW estimator has seen limited uptake in practice: the applications we are aware of are a simulation comparison of imputation variance estimators with accompanying code (Hughes et al., 2016) and the EHR validation analyses of GS and colleagues (Giganti et al., 2020; Giganti and Shepherd, 2020). A central obstacle is that, for a long time, no general software implemented the method; Giganti and Shepherd (2020) made RW more accessible by providing implementation code and demonstrating it on real EHR data, but only for a single parametric imputation model and a limited set of settings. Modern imputation workflows raise two further challenges. First, most analysts impute by multiple imputation with chained equations (MICE), in which each variable is imputed from a separate conditional model with no joint imputation likelihood (van Buuren and Groothuis-Oudshoorn, 2011), so the exact estimator RW proposed cannot be applied directly. Second, predictive mean matching (PMM)—the default method in MICE for continuous variables—is not a parametric model: the imputed value is copied from a matched observed donor rather than drawn from a parametric distribution (Schenker and Taylor, 1996; Andridge and Little, 2010), whereas RW was developed for parametric imputation.

This paper addresses these challenges and makes three contributions. First, we develop an open-source R package, `rw`, implementing the RW variance estimator; the package integrates with `mice` output, and the syntax for RW variance estimation mimics that of `mice` for RR variance estimation. Second, we generalize the RW variance estimator to chained conditional (MICE) imputation by stacking the score and nuisance contributions from each conditional model block, with no joint imputation likelihood, so that the package can be used with any parametric imputation method in `mice`. Third, we extend the RW variance estimator to PMM, a form of nonparametric imputation for which no RW-type variance estimator has previously been proposed; related variance ideas for hot-deck and nearest-neighbor imputation have been studied in survey sampling (Rao and Shao, 1992; Skinner, 2002; Kim and Fuller, 2004; Yang and Kim, 2020) but have not been integrated into the RW framework. Throughout, our focus is on variance calibration for existing imputation workflows.

In Section 2, we describe the data structure and notation, the MI point estimator, and the RW variance estimator. We develop extensions to MICE and PMM in Section 3. In Section 4, we describe the software implementation, and we evaluate our proposal compared with RR in the original RW missing-data benchmark and in the GS measurement-error validation setting in Section 5. We conclude by discussing limitations and directions of future research in Section 6.

## 2 Variance estimation with imputed data

### 2.1 Data structure and notation

We observe independent and identically distributed data  $O_1, \dots, O_n$ , where each  $O_i$  records the components that are observed for subject  $i$  together with its missingness pattern. We write  $Z_i$  for the complete data of subject  $i$ : the ideal data we would wish to observe on every subject, comprising both the observed and the missing components. Throughout,  $P_n$  denotes the empirical average,  $P_n f = n^{-1} \sum_{i=1}^n f_i$ . The parameter of interest is a finite-dimensional parameter  $\beta \in \mathbb{R}^q$  defined through a complete-data estimating function  $U(\cdot; \beta)$  as the unique solution of the population estimating equation  $E\{U(Z; \beta_0)\} = 0$ . This formulation is general:  $\beta$  need not be a regression coefficient, but can be any parameter admitting an unbiased complete-data estimating function. We call  $\beta_0$  the data-generating value of the parameter. When  $Z$  is partially missing the estimating function cannot be evaluated directly, and multiple imputation replaces the missing components by draws from an imputation model. In this section we consider parametric imputation, the setting in which the RW estimator was developed: we index the imputation model by a parameter  $\psi \in \mathbb{R}^c$ , estimated from the observed data by  $\hat{\psi}$  solving an observed-data score equation  $\sum_{i=1}^n S_{\text{obs},i}(\psi) = 0$ , with probability limit  $\psi^*$ . Predictive mean matching (PMM), in contrast, copies each missing value from an observed donor with a similar predicted mean rather than drawing it from a fitted distribution (Schenker and Taylor, 1996; Andridge and Little, 2010); we defer PMM, and the additional donor-index notation it requires, to Section 3.2. The influence function of  $\hat{\psi}$  is the mean-zero function  $d_i$  satisfying  $\hat{\psi} - \psi^* = n^{-1} \sum_{i=1}^n d_i + o_p(n^{-1/2})$ ; it summarizes how much subject  $i$ 's observed data move the imputation-parameter estimate. Drawing from the fitted imputation model  $m$  times produces  $m$  completed datasets, indexed by  $p = 1, \dots, m$ . We write  $Z_i^{(p)}$  for the completed-data row of subject  $i$  in imputation  $p$  and  $U_i^{(p)}(\beta) = U\{Z_i^{(p)}; \beta\}$  for the corresponding complete-data estimating-function contribution.

### 2.2 Estimation using multiple imputation

Given the  $m$  completed datasets, the multiply imputed estimator  $\hat{\beta}$  averages the completed-data estimating functions across imputations and solves the resulting equation,

$$0 = P_n \bar{U}(\hat{\beta}) = \frac{1}{n} \sum_{i=1}^n \bar{U}_i(\hat{\beta}), \quad \bar{U}_i(\beta) = \frac{1}{m} \sum_{p=1}^m U_i^{(p)}(\beta).$$

Equivalently, one may solve the complete-data estimating equation within each imputation and average the solutions; the two agree to first order, and we use the averaged-score form throughout. Two population parameters are of potential interest. The first is the data-generating value  $\beta_0$  of Section 2.1. The second is the probability limit of  $\hat{\beta}$  itself, denoted  $\beta^*$ : the value  $\hat{\beta}$  converges to as  $n \rightarrow \infty$  under the implemented imputation and analysis procedure, with the imputation model held fixed at its own limit  $\psi^*$ . By construction  $\hat{\beta}$  is consistent for  $\beta^*$  whether or not the imputation model is correct, so  $\beta^*$  is a target on which we can always hope to make valid inference. The two coincide, i.e.,  $\beta^* = \beta_0$ , precisely when the implemented procedure leaves the estimating equation unbiased at  $\beta_0$ , and they can diverge through two distinct routes. The first route is misspecification of the imputation model. Misspecification

can shift  $\beta^*$  away from  $\beta_0$  but need not do so: what matters is whether the features of the conditional distribution that enter the analysis score are correctly reproduced. In the simulations of Section 5, for example, imputation models with a misspecified error variance still yield essentially unbiased point estimates, whereas a misspecified mean structure or outcome-dependent observation shifts  $\beta^*$  away from  $\beta_0$ . The second route does not require misspecification. If the analysis evaluates a post-imputation quantity whose definition depends on the imputed values, as when eligibility for the analysis subset is determined by an imputed variable, then  $\beta^*$  is defined through the imputed-data eligibility rule and need not equal  $\beta_0$ , even when the imputation model is correctly specified. This is the situation of the GS validation setting studied in Section 5. Whether  $\beta^*$  can be interpreted as the scientific target also depends on the missingness mechanism. Under missingness completely at random (MCAR), a complete-case analysis is already unbiased for  $\beta_0$ , and imputation serves mainly to recover efficiency. Under missingness at random (MAR) given the observed data, a correctly specified imputation model yields  $\beta^* = \beta_0$ , whereas complete-case analysis may be biased. Under missingness not at random (MNAR),  $\beta^*$  generally differs from  $\beta_0$  for any imputation model fitted to the observed data alone, absent additional assumptions; see Little et al. (2024) for a comparison of complete-case analysis, inverse probability weighting, and multiple imputation across these mechanisms. Variance calibration, by contrast, is agnostic to the mechanism: coverage of  $\beta^*$  concerns only the sampling spread of  $\hat{\beta}$ , so the diagnostics developed in this paper apply under any mechanism. This distinction matters for assessing variance calibration. Coverage of  $\beta^*$  is a pure variance diagnostic: it asks whether the estimated standard error correctly captures the sampling spread of  $\hat{\beta}$ , whatever that estimator targets and under any missingness mechanism. Coverage of  $\beta_0$  additionally reflects any systematic departure of  $\beta^*$  from  $\beta_0$ , so it conflates variance calibration with potential bias of the implemented point estimator.

### 2.3 Variance estimation using Rubin’s rules

The standard approach estimates the variance of  $\hat{\beta}$  by Rubin’s rules (RR). Let  $\hat{\beta}^{(p)}$  be the estimate obtained from completed dataset  $p$ , so that  $\hat{\beta} = m^{-1} \sum_{p=1}^m \hat{\beta}^{(p)}$ , and let  $\widehat{W}^{(p)}$  be the corresponding complete-data (model-based) variance estimate of  $\hat{\beta}^{(p)}$ . RR combines a within-imputation and a between-imputation component,

$$\bar{W} = \frac{1}{m} \sum_{p=1}^m \widehat{W}^{(p)}, \quad \hat{B} = \frac{1}{m-1} \sum_{p=1}^m (\hat{\beta}^{(p)} - \hat{\beta})(\hat{\beta}^{(p)} - \hat{\beta})^\top,$$

into the total-variance estimator

$$\hat{V}_{\text{RR}} = \bar{W} + \left(1 + \frac{1}{m}\right) \hat{B}.$$

The within-imputation term  $\bar{W}$  averages the complete-data sampling variability across imputations, and the between-imputation term  $\hat{B}$  measures how much the point estimate moves from one imputed dataset to another; the factor  $1 + 1/m$  corrects for using finitely many imputations. RR requires only the per-imputation estimates and their model-based variances, which is what makes it simple to apply. Its validity, however, rests on congeniality between the imputation and analysis procedures (Rubin, 1987; Meng, 1994; Wang and Robins, 1998): when the two are congenial,  $\hat{V}_{\text{RR}}$  is consistent for the repeated-sampling variance of  $\hat{\beta}$ ; when they are not, it can

be biased in either direction. Figure 1 illustrates the two ways RR can fail, alongside the role of  $\beta^*$ ; the three panels span the three regimes identified above: congeniality, anti-conservative uncongeniality, and conservative uncongeniality. In panel (a) the imputation and analysis procedures are congenial and both correctly specified:  $\beta^* = \beta_0$ , the RR formula matches the true sampling spread, and both RR and RW confidence intervals achieve approximately 95% coverage of  $\beta^*$ . In panel (b) congeniality fails in the *anti-conservative* direction: the analysis exploits structure that the imputation model does not reflect, so the between-imputation component understates the true variability, and the RR interval is too narrow. In panel (c) congeniality fails in the *conservative* direction: the imputation model uses information beyond that of the analysis, inflating the between-imputation component and producing an interval that is too wide. Panel (c) also shows the bias case  $\beta^* \neq \beta_0$ : coverage of  $\beta^*$  overshoots 95% under RR and is near 95% under RW, but coverage of  $\beta_0$  depends additionally on the distance between  $\beta^*$  and  $\beta_0$ . Bias is governed by the correctness of the imputation procedure rather than by congeniality and could in principle arise in any panel; we display it once, in panel (c), because this combination, conservative uncongeniality together with a shifted target, is the one that arises in the GS validation setting of Section 5. In all three panels the RW estimator correctly sizes the interval relative to the true sampling spread, producing near-nominal coverage of  $\beta^*$ .

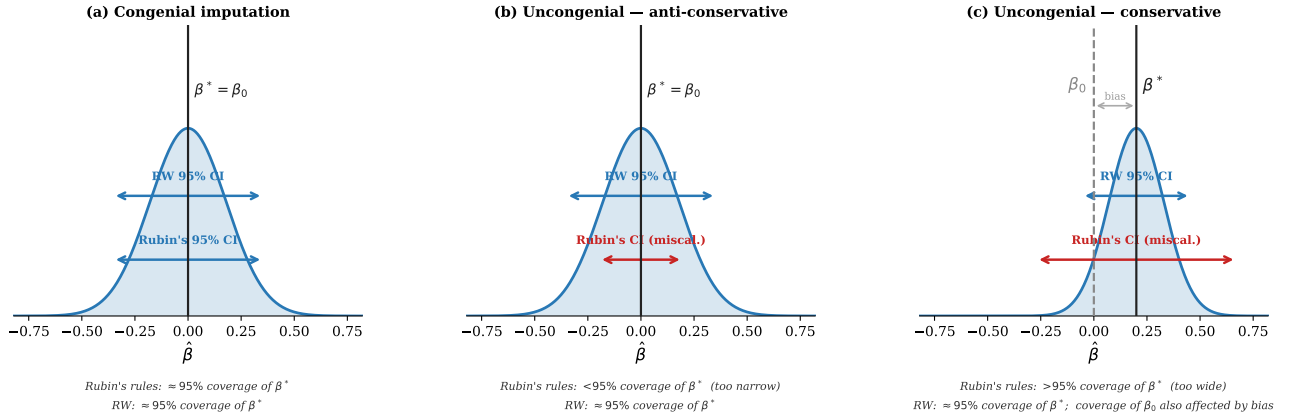


Figure 1: Conceptual illustration of variance calibration for multiply imputed estimators. Each panel shows the sampling distribution of  $\hat{\beta}$  (shaded curve) under repeated study replications, the probability limit  $\beta^*$  (solid vertical line), and, in panel (c), the data-generating coefficient  $\beta_0$  (dashed vertical line). Horizontal brackets indicate 95% confidence intervals from RR (lower bracket, red when miscalibrated) and from the RW estimator (upper bracket, blue). Panel (a): congenial imputation, both intervals are correctly sized and achieve approximately 95% coverage of  $\beta^*$ . Panel (b): uncongenial, anti-conservative case, the RR interval is too narrow, leading to under-coverage of  $\beta^*$ ; the RW interval is correctly sized. Panel (c): uncongenial, conservative case with estimator bias ( $\beta^* \neq \beta_0$ ), the RR interval is too wide, producing over-coverage of  $\beta^*$ ; coverage of  $\beta_0$  depends additionally on the bias  $\beta^* - \beta_0$ . The primary simulation diagnostic in Section 5 is coverage of  $\beta^*$ , because it isolates variance calibration from point-estimator bias.

## 2.4 Variance estimation using Robins–Wang

### 2.4.1 The three sources of variability

We begin by giving an overview of each component of the Robins–Wang (RW) variance estimator, and then provide formal definitions. The multiply imputed estimator  $\hat{\beta}$  solves an averaged complete-data estimating equation, and its repeated-sampling variability has exactly three first-order sources. *Source 1: analysis-score variability.* Even if the imputation-model parameter  $\psi$  were known exactly and held fixed,  $\hat{\beta}$  would still vary across repeated study replications, because the analysis scores  $U_i(\beta)$  fluctuate with the observed data. This is the same variability that ordinary complete-data sandwich inference would estimate. It does not vanish as the number of imputations  $m$  grows, because it reflects between-subject variation in the analysis, not between-imputation variation. *Source 2: imputation-parameter uncertainty.* In practice,  $\psi$  is estimated from the observed data, and this estimation uncertainty propagates into  $\hat{\beta}$  through the imputed values. The propagation has two steps. First,  $\hat{\psi}$  fluctuates around its limit across replications, with magnitude governed by its influence function  $d_i$  (Section 2.1). Second, those fluctuations shift the imputed values, which in turn shift the analysis score; how strongly depends on the sensitivity of the analysis score to the imputation score. These quantities are defined through the parametric imputation score; Section 3.2 shows how a smooth donor-selection working law supplies the analogous score for PMM. *Source 3: cross-covariance.* Sources 1 and 2 are not independent, because the same observed records drive both the analysis score and the imputation-parameter estimator. A subject that pushes  $\hat{\psi}$  in one direction also tends to shift the analysis score in a correlated direction. Ignoring this dependence would misrepresent the true variance, so a cross-covariance term accounts for it explicitly. How does RR compare? The within-imputation variance is an estimate of Source 1, and the between-imputation variance is an approximation of Source 2. However, their additive combination is valid only under congeniality. When the imputation and analysis procedures are uncongenial, the between-imputation component conflates Sources 2 and 3 incorrectly: it inflates the variance in settings where the imputation model uses information beyond the analysis (as in the GS validation benchmark, where RR was conservative by 29%), and deflates it where the analysis exploits structure the imputation model does not capture (as in the RW missing-data benchmark, where RR is anti-conservative). The RW estimator builds all three sources directly from estimating-equation quantities, which is what makes it valid without congeniality. The formal definitions below make these three sources precise.

### 2.4.2 Components of the RW variance estimator

We first present the RW variance estimator in the setting RW originally considered, in which all missing values are drawn from a parametric imputation model indexed by  $\psi$ ; in Section 3 we extend the RW variance estimator to chained-equation and PMM imputation. Beyond the influence function  $d_i$  of  $\hat{\psi}$  introduced in Section 2.1, one further score-type object enters the RW calculation: the missing-value score  $S_{\text{mis},i}$ , the imputation-model score evaluated at the imputed value for record  $i$ . In sample form the influence function is

$$d_i = -\left\{P_n \partial S_{\text{obs},i} / \partial \psi^\top\right\}^{-1} S_{\text{obs},i}.$$

Because  $S_{\text{obs},i}$  is zero for records with nothing observed for the imputed variable,  $d_i$  is supported only on records that contribute to the imputation fit, and the sums defining  $\hat{\alpha}$  and  $\hat{C}$  below

run effectively over those records. Robins and Wang (2000) express the first-order variance of  $\hat{\beta}$  using  $U$ ,  $S_{\text{mis}}$ , and  $d$ . In sample form, the sandwich variance estimator is

$$\hat{\Gamma} = \frac{1}{n} \hat{\tau}^{-1} \hat{\Delta} (\hat{\tau}^{-1})^\top, \quad \hat{\Delta} = \hat{\Omega} + \hat{\kappa} \hat{\alpha} \hat{\kappa}^\top + \hat{C}.$$

The three matrices inside  $\hat{\Delta}$  are empirical second moments of order one; the leading  $1/n$  converts the sandwich limit for  $\sqrt{n}(\hat{\beta} - \beta^*)$  into the finite-sample variance of  $\hat{\beta}$ . We keep this scaling explicit so that  $\hat{\Delta}$  is directly comparable with the population formulas in Robins and Wang (2000). The *bread*  $\hat{\tau}$  is the average derivative of the completed-data estimating equation with respect to  $\beta$ ,

$$\hat{\tau} = \frac{1}{nm} \sum_{p=1}^m \sum_{i=1}^n \left. \frac{\partial U_i^{(p)}(\beta)}{\partial \beta^\top} \right|_{\beta=\hat{\beta}}.$$

It linearises the estimating equation around the solution  $\hat{\beta}$ , converting the sandwich into units of  $\text{Var}(\hat{\beta})$ . We use *bread* and *meat* in their standard sandwich-variance sense:  $\hat{\tau}$  estimates the expected derivative of the averaged estimating function, and  $\hat{\Delta}$  estimates the variance of its subject-level contributions. As described above, the *meat*  $\hat{\Delta}$  is made up of three components, corresponding to the three sources of variance. We now give their explicit forms. *Source 1: analysis-score covariance.* The first piece is the outer product of each subject's imputation-averaged analysis score  $\bar{U}_i$ . Averaging over imputations within each subject eliminates imputation noise, leaving only the between-subject variation in the analysis score that would exist even if  $\psi$  were known:

$$\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n \bar{U}_i(\hat{\beta}) \bar{U}_i(\hat{\beta})^\top.$$

*Source 2: imputation-parameter uncertainty.* The second piece is assembled from two building blocks. The matrix  $\hat{\kappa}$  measures how sensitively the analysis score responds to a change in the imputed value: it is the sample cross-moment of the analysis score  $U_i^{(p)}$  and the missing-value score  $S_{\text{mis},i}^{(p)}$ , which is the gradient of the imputation log-likelihood with respect to  $\psi$  evaluated at the imputed value. A large  $\hat{\kappa}$  means that small shifts in  $\psi$ , and hence in the imputed values, substantially move the analysis score. The matrix  $\hat{\alpha}$  is the variance of the influence contribution  $d_i^{(p)}$  across subjects and imputations; it measures how much  $\hat{\psi}$  fluctuates as the observed data change. Together,

$$\hat{\kappa} = \frac{1}{nm} \sum_{p=1}^m \sum_{i=1}^n U_i^{(p)}(\hat{\beta}) S_{\text{mis},i}^{(p)\top}, \quad \hat{\alpha} = \frac{1}{nm} \sum_{p=1}^m \sum_{i=1}^n d_i^{(p)} d_i^{(p)\top},$$

and the contribution of imputation-parameter estimation uncertainty to  $\text{Var}(\hat{\beta})$  is  $\hat{\kappa} \hat{\alpha} \hat{\kappa}^\top$ . *Source 3: cross-covariance.* The third piece corrects for the dependence between Sources 1 and 2. Because subject  $i$  simultaneously contributes to  $\bar{U}_i$  (the analysis score) and to  $\bar{d}_i$  (the influence on  $\hat{\psi}$ ), the two sources are correlated at the individual level. Writing  $\bar{d}_i = m^{-1} \sum_{p=1}^m d_i^{(p)}$  for the imputation-averaged influence contribution, the cross-covariance is

$$\hat{C} = \frac{1}{n} \left[ \hat{\kappa} \left\{ \sum_{i=1}^n \bar{d}_i \bar{U}_i^\top \right\} + \left\{ \sum_{i=1}^n \bar{U}_i \bar{d}_i^\top \right\} \hat{\kappa}^\top \right] = \frac{1}{n} [\hat{\kappa} \bar{d}^\top \bar{U} + \bar{U}^\top \bar{d} \hat{\kappa}^\top],$$



where  $\bar{U}$  and  $\bar{d}$  are the  $n \times q$  and  $n \times c$  matrices with rows  $\bar{U}_i^\top$  and  $\bar{d}_i^\top$ , and  $c = \dim(\psi)$  as introduced in Section 2.1. The symmetrisation (adding the transpose) ensures  $\hat{\Delta}$  is symmetric. The RR variance estimator implicitly sets a version of this term to zero, which is one reason it is miscalibrated under uncongeniality.

### 3 Improved variance estimation with multiply-imputed data

This section extends the RW framework beyond the parametric imputation model assumed in Section 2.4. For chained conditional (MICE) imputation, there is no joint imputation score:  $S_{\text{mis}}$  and  $d$  are replaced by stacked vectors assembled from each conditional model block, as developed in Section 3.1; we call the resulting estimator RW-MICE. For predictive mean matching, the completed value is copied from an observed donor rather than drawn from a parametric distribution, so two things change: the donor selection needs a differentiable working score, and completed rows sharing a donor are not independent summands, requiring  $\hat{\Omega}$  and  $\hat{C}$  to be formed at the donor-source level. This is developed in Section 3.2, yielding the RW-PMM estimator.

#### 3.1 RW variance for chained conditional imputation models

The RW framework of Section 2.4 requires three ingredients tied to a joint parametric imputation likelihood: a parameter vector  $\psi$  indexing the joint imputation distribution, an observed data score equation  $\sum_i S_{\text{obs},i}(\psi) = 0$  whose solution defines  $\hat{\psi}$ , and the resulting missing value score  $S_{\text{mis},i}$  and nuisance influence  $d_i$ . MICE supplies none of these. Its conditional working models are fitted block by block, where a block is a single variable imputed by its own conditional model (the default in `mice`) or a group of variables imputed jointly; the blockwise fits do not correspond to any joint imputation likelihood, and admit no global parameter solving a single score equation. Even if one concatenated the blockwise parameters as  $\psi = (\psi_1, \dots, \psi_R)$ , where  $R$  denotes the number of imputation blocks, MICE conditional models are in general mutually incompatible: they need not correspond to any joint distribution over the missing variables, so no joint likelihood exists and no global score equation of the form  $\sum_i S_{\text{obs},i}(\psi) = 0$  can be defined. Our proposed resolution is to shift from one global estimated parameter to one realized parameter draw per block per imputation. In each imputation  $p$ , MICE generates a parameter state  $\psi_r^{(p)}$  for block  $r$ , the coefficient draw used to produce that block’s completed values, and this realized state is already recorded in a standard MICE run in `mice`. We evaluate the RW score and nuisance components at these realized states, one block at a time, and stack the per-block contributions into composite vectors that are substituted into the unchanged RW formula. No joint imputation likelihood is assumed, and no modeling choices are required beyond the conditional models that MICE already fits. The non-trivial step is justifying that stacking works. Two conditions must hold: the realized states  $\psi_r^{(p)}$  must concentrate around their blockwise working limits  $\psi_{r0}$  fast enough that evaluating at  $\psi_r^{(p)}$  rather than  $\psi_{r0}$  changes the empirical moments only at negligible order; and the stacked empirical moments must consistently estimate the population RW variance components for the MICE estimator without a coherent joint likelihood. Both are established in Appendix A; the rest of this section builds the construction.

### 3.1.1 Sequential conditional working law

Suppose the missing variables are imputed in blocks  $r = 1, \dots, R$ . For block  $r$ , let  $H_{r,i}^{(p)}$  denote the predictors and previously updated variables used by the conditional model in completed dataset  $p$ , with working model  $f_r\{Y_{r,i} \mid H_{r,i}^{(p)}; \psi_r\}$ . Because the conditional models need not be mutually compatible, each block has its own working parameter  $\psi_r$ , whose working value  $\psi_{r0}$  is defined as the probability limit of the corresponding conditional estimator, rather than as a coordinate of a global  $\psi$ . There is one  $\psi_r$  per block and no relationship between the  $\psi_r$  values across blocks is assumed or needed. In a proper MICE run, imputation  $p$  draws a parameter state  $\psi_r^{(p)}$  for each block and generates the completed block from it. These realized states are already stored as part of the imputation output in `mice`. We evaluate all RW score components at these realized states  $\psi_r^{(p)}$ . They are the natural plug-in: already produced by the MICE run, requiring no additional estimation, and under standard regularity conditions for proper imputation they concentrate around the blockwise working limits at rate  $n^{-1/2}$ ,

$$\psi_r^{(p)} - \psi_{r0} = O_p(n^{-1/2}).$$

Because the RW score components are smooth functions of the imputation parameter, a first-order Taylor expansion shows that evaluating at  $\psi_r^{(p)}$  rather than  $\psi_{r0}$  changes the empirical RW moments only at order  $n^{-1/2}$ , which is higher order than the leading term that determines the sandwich limit. The substitution error is therefore asymptotically negligible. The formal statement is in Appendix A. Next, we describe the score and nuisance objects required for the RW variance computation. Write the conditional-model score contribution for block  $r$ , subject  $i$ , imputation  $p$  as

$$s_{r,i}^{(p)}(\psi_r) = \frac{\partial}{\partial \psi_r} \log f_r\{Y_{r,i} \mid H_{r,i}^{(p)}; \psi_r\}.$$

The missing-value score for rows imputed in block  $r$  is the working score evaluated at the realized state,  $S_{r,\text{mis},i}^{(p)} = s_{r,i}^{(p)}(\psi_r^{(p)})$ ; for rows not imputed in block  $r$ , it is zero. A large score means a small change in  $\psi_r$  would have moved the imputed value substantially. The nuisance influence is constructed as follows. Let  $\delta_{r,i} = 1$  indicate that block  $r$  is observed for subject  $i$ , and set  $S_{r,\text{obs},i}^{(p)} = \delta_{r,i} s_{r,i}^{(p)}(\psi_r^{(p)})$ . Define

$$A_r^{(p)} = P_n \left\{ \frac{\partial s_{r,i}^{(p)}(\psi_r)}{\partial \psi_r^\top} \bigg|_{\psi_r = \psi_r^{(p)}} \delta_{r,i} \right\},$$

the observed-data Hessian of the block- $r$  conditional log-likelihood. The blockwise nuisance influence is the Newton linearization

$$d_{r,i}^{(p)} = -\{A_r^{(p)}\}^{-1} S_{r,\text{obs},i}^{(p)},$$

the block- $r$  analogue of the single-model  $d_i$ : it measures how much the block- $r$  parameter draw would shift if subject  $i$ 's data changed. It requires no additional modeling beyond the conditional model that MICE already fits. Like the single-model case,  $d_{r,i}^{(p)}$  is supported only on subjects with block  $r$  observed. Stacking across all  $R$  blocks then gives the MICE imputation score and nuisance influence vectors for subject  $i$  in imputation  $p$ :

$$S_{\text{mis},i}^{(p)} = (S_{1,\text{mis},i}^{(p)\top}, \dots, S_{R,\text{mis},i}^{(p)\top})^\top, \quad d_i^{(p)} = (d_{1,i}^{(p)\top}, \dots, d_{R,i}^{(p)\top})^\top.$$

The conditions required of each block are minimal: a regular observed-score derivative (so that  $A_r^{(p)}$  is invertible), finite second moments for the score contributions, and a realized state that concentrates at its working limit. No coherent joint likelihood across blocks is needed. Stacking automatically captures cross-block parameter uncertainty. The off-diagonal elements of  $\alpha = E[d_i d_i^\top]$ , such as  $E[d_{r,i} d_{r',i}^\top]$  for  $r \neq r'$ , reflect correlation between estimation uncertainty in different blocks, which arises because the same observed data contribute to both conditional model fits. These terms are non-zero in general and are automatically included when  $\hat{\alpha}$  is formed from the full stacked vectors  $d_i^{(p)}$ .

### 3.1.2 RW components for the MICE estimator

The analysis estimator is still defined by the averaged completed-data estimating equation  $0 = P_n \bar{U}(\hat{\beta})$ . All RW components are therefore similar to their Section 2.4 counterparts, with the single-model  $S_{\text{mis},i}^{(p)}$  and  $d_i^{(p)}$  replaced by the stacked MICE quantities defined above; we call the resulting variance estimator RW-MICE. Concretely, the bread  $\hat{\tau}$  is computed exactly as before; the three covariance pieces are:

$$\begin{aligned}\hat{\Omega} &= \frac{1}{n} \sum_{i=1}^n \bar{U}_i(\hat{\beta}) \bar{U}_i(\hat{\beta})^\top, \\ \hat{\kappa} &= \frac{1}{nm} \sum_{p=1}^m \sum_{i=1}^n U_i^{(p)}(\hat{\beta}) S_{\text{mis},i}^{(p)\top}, \quad \hat{\alpha} = \frac{1}{nm} \sum_{p=1}^m \sum_{i=1}^n d_i^{(p)} d_i^{(p)\top}, \\ \hat{C} &= \frac{1}{n} [\hat{\kappa} \bar{d}^\top \bar{U} + \bar{U}^\top \bar{d} \hat{\kappa}^\top],\end{aligned}$$

where  $\bar{U}$  and  $\bar{d}$  are the  $n \times q$  and  $n \times \dim(\psi)$  matrices with rows  $\bar{U}_i^\top$  and  $\bar{d}_i^\top$ , and  $\dim(\psi) = \sum_r \dim(\psi_r)$  is the total number of imputation parameters across all blocks. Stacking changes the internal block structure of the matrices. The matrix  $\hat{\kappa} = [\hat{\kappa}_1, \dots, \hat{\kappa}_R]$  is block-partitioned by imputation block, with  $\hat{\kappa}_r = (nm)^{-1} \sum_{p,i} U_i^{(p)} S_{r,\text{mis},i}^{(p)\top}$  measuring how sensitive the analysis score is to block  $r$ 's parameter; because the analysis equation mixes contributions from all imputed variables,  $\hat{\kappa}_r$  is typically non-zero even for blocks whose variables do not appear directly in the analysis estimating function (in parametric regression analysis models, blocks other than the primary analysis covariate). The matrix  $\hat{\alpha}$  has  $(r, r')$  block  $\hat{\alpha}_{rr'} = (nm)^{-1} \sum_{p,i} d_{r,i}^{(p)} d_{r',i}^{(p)\top}$ ; the off-diagonal blocks for  $r \neq r'$  capture cross-block parameter uncertainty and are non-zero whenever subjects contribute to both conditional model fits. Dropping them would incorrectly assume independence of estimation uncertainty across blocks, which fails unless missing data patterns are block-monotone with no shared covariates. Consistency of the stacked estimator is established in Appendix A; when  $R = 1$  the construction reduces to the single-model RW estimator of Section 2.4. The MICE extension requires no new modeling beyond what a standard MICE run already produces. Given the completed datasets and the recorded parameter draws  $\psi_r^{(p)}$ , every RW component is computed from formulas unchanged in form from Section 2.4. The only structural change is that  $S_{\text{mis}}$  and  $d$  are block vectors, and the resulting sandwich automatically accounts for within-block and cross-block parameter uncertainty through the full block structure of  $\hat{\alpha}$ .

## 3.2 RW variance estimation with predictive mean matching

Predictive mean matching (PMM) is attractive in medical applications because it imputes plausible observed values rather than unrestricted model-based draws: the imputed value is not a prediction but a copy of an actual observed record, which respects distributional constraints that a parametric draw might violate. The same feature, however, creates two problems for the RW variance construction that do not arise in fully parametric imputation.

The first problem concerns differentiation of the donor-selection rule. Standard PMM selects the donor by finding the  $k$  observed records whose predictive mean is closest to the recipient's predictive mean, the donor pool, and sampling one uniformly from that pool. This nearest-neighbor rule is a step function of the imputation parameter  $\psi$ : a small shift in  $\psi$  that does not change the donor pool produces no change in the donor probabilities, while a shift that just excludes a current neighbor produces a discontinuous jump. A direct log-gradient of the selected donor's probability is therefore zero wherever the pool is unchanged and undefined at a boundary. It does not provide a regular missing-value score for the RW cross-moment  $\kappa = E\{US_{\text{mis}}^\top\}$ . This difficulty does not prevent calculation of the fitted regression's influence contributions  $d_i$  or their second moment  $\alpha$ .

The second problem is that PMM induces correlation between recipients that share a donor. When several missing recipients copy from the same observed donor, their completed-data analysis scores share a common stochastic source, the value of the donor record (Rao and Shao, 1992; Skinner, 2002). A random shift in the donor's observed value can affect several recipient scores simultaneously. The rowwise second moment  $\sum_i U_i U_i^\top$  does not include cross-products between these records. Their omission need not always decrease the estimated variance; the direction depends on the analysis scores.

In this section, we propose an RW estimator that addresses each of these problems at the specific variance component it affects; we call this estimator RW-PMM, to contrast with the RW-MICE estimator of Section 3.1. Section 3.2.1 defines a differentiable working probability for calculating  $\hat{\kappa}$ , without changing the imputations produced by `mice`. Section 3.2.3 aggregates analysis scores by their recorded donor to construct  $\hat{\Omega}$  and  $\hat{C}$ . These are working approximations for variance estimation, not changes to the PMM sampling rule.

### 3.2.1 Donor probabilities for variance estimation

Consider one continuous variable  $A$  imputed by PMM, with observed and missing sets  $\mathcal{O}_A$  and  $\mathcal{M}_A$ . Let  $x_i \in \mathbb{R}^c$  contain the fully observed matching predictors, including an intercept. Under the default matching rule in `mice`, observed donors use the fitted regression coefficient  $\hat{\psi}^{(p)}$ , whereas recipients use a coefficient draw  $\dot{\psi}^{(p)}$ . Their matching predictions are

$$a_j^{(p)} = x_j^\top \hat{\psi}^{(p)}, \quad t_i^{(p)} = x_i^\top \dot{\psi}^{(p)}.$$

Let  $\mathcal{N}_i^{(p)}$  contain the  $k$  donor predictions closest to  $t_i^{(p)}$ . The actual imputation is unchanged:

$$P\{J_i^{(p)} = j \mid \text{fitted model and coefficient draw}\} = \frac{\mathbf{1}\{j \in \mathcal{N}_i^{(p)}\}}{k}, \quad A_i^{(p)} = A_{J_i^{(p)}}^{\text{obs}}.$$

The selected donor IDs and completed values are retained exactly.

For the variance calculation, suppress the superscript  $p$  temporarily and order the  $N = |\mathcal{O}_A|$  donor predictions as  $a_{(1)} < \dots < a_{(N)}$ . The following derivatives assume distinct predictions and

a locally unchanged ordering. Donor  $(r)$  is among the  $k$  nearest donors to a scalar prediction  $t$  precisely when  $\ell_{(r)} < t < u_{(r)}$ , apart from ties at the endpoints, where

$$\ell_{(r)} = \begin{cases} -\infty, & r \leq k, \\ (a_{(r-k)} + a_{(r)})/2, & r > k, \end{cases} \quad u_{(r)} = \begin{cases} (a_{(r)} + a_{(r+k)})/2, & r + k \leq N, \\ \infty, & r + k > N. \end{cases}$$

These midpoints are the locations at which a competing donor enters or leaves the nearest  $k$  positions.

Define an auxiliary prediction  $T_i = t_i + s_i Z$ , where  $Z \sim N(0, 1)$  and

$$s_i^2 = x_i^\top \Sigma x_i, \quad \Sigma = \sigma^2 V.$$

Here  $\sigma$  is the residual standard-deviation draw from the PMM regression, and  $V$  is its fitted coefficient covariance factor. For a full-rank unregularized fit,  $V = (X_{\mathcal{O}_A}^\top X_{\mathcal{O}_A})^{-1}$ ; the implementation uses the factor from the `mice` regression fit. Integrating the uniform top- $k$  selection rule over this auxiliary normal distribution gives the working donor probability

$$p_{ij} = \frac{1}{k} \left\{ \Phi\left(\frac{u_j - t_i}{s_i}\right) - \Phi\left(\frac{\ell_j - t_i}{s_i}\right) \right\}. \quad (1)$$

The expression in braces is the probability of entering the donor pool; division by  $k$  gives the probability of selection. Thus  $\sum_{j \in \mathcal{O}_A} p_{ij} = 1$ . No auxiliary prediction is drawn when imputing the data, and the recorded donors are not sampled again from  $p_{ij}$ . Equation (1) is a normal working approximation for variance estimation, not the conditional sampling probability of the recorded `mice` donor. Its smoothing scale is determined by the fitted regression covariance; no additional bandwidth is selected to target a particular coverage level.

To differentiate, shift both  $\hat{\psi}$  and  $\dot{\psi}$  by the same vector  $\delta$ , holding their difference and  $\Sigma$  fixed. This allows both donor predictions and the recipient prediction to move. For finite endpoints, their derivatives are

$$\dot{\ell}_{(r)} = (x_{(r-k)} + x_{(r)})/2, \quad \dot{u}_{(r)} = (x_{(r)} + x_{(r+k)})/2.$$

Writing  $z_{ij}^u = (u_j - t_i)/s_i$  and  $z_{ij}^\ell = (\ell_j - t_i)/s_i$ , differentiation gives

$$\dot{p}_{ij} := \nabla_\delta p_{ij}(\delta)|_{\delta=0} = \frac{\phi(z_{ij}^u)(\dot{u}_j - x_i) - \phi(z_{ij}^\ell)(\dot{\ell}_j - x_i)}{k s_i}. \quad (2)$$

Terms corresponding to infinite endpoints are zero. The derivatives sum to zero over donors. An intercept shift moves all predictions equally, so its derivative is zero; this does not force the derivatives for the other coefficients to vanish. Holding  $\Sigma$  fixed is part of the working linearization, not a claim that its estimation adds no uncertainty.

### 3.2.2 Expected analysis scores and the PMM cross-moment

The RW cross-moment involves the analysis score as well as the imputation score. Under the working donor law, define  $S_{ij} = \dot{p}_{ij}/p_{ij}$  for  $p_{ij} > 0$ . Let  $G_{ij}^{(p)}$  be a working expectation of recipient  $i$ 's analysis score when donor  $j$  is considered. Then

$$\sum_{j \in \mathcal{O}_A} p_{ij}^{(p)} G_{ij}^{(p)} S_{ij}^{(p)\top} = \sum_{j \in \mathcal{O}_A} G_{ij}^{(p)} \dot{p}_{ij}^{(p)\top}.$$

This identity motivates averaging over candidate donors when calculating the PMM columns of  $\kappa$ , rather than multiplying the realized recipient score by the working score of its selected donor:

$$\widehat{\kappa}_{\text{PMM}} = \frac{1}{nm} \sum_{p=1}^m \sum_{i \in \mathcal{M}_A} \sum_{j \in \mathcal{O}_A} G_{ij}^{(p)} \dot{p}_{ij}^{(p)\top}. \quad (3)$$

Only the donor probabilities are differentiated here; the calculation does not add a term involving  $\partial G_{ij}/\partial \psi$ . The identity is exact under the working law, but its use for imputations from `mice` remains an approximation.

To compute  $G_{ij}^{(p)}$ , the donor value is integrated under the fitted normal working model  $A_j^\dagger \sim N(a_j^{(p)}, (\sigma^{(p)})^2)$ , rather than fixed at its observed value. The recipient's own observed covariates are retained. If a subsequent variable is imputed using  $A$  as a predictor, its distribution is evaluated at  $A_j^\dagger$  using the recorded conditional model. In notation,

$$G_{ij}^{(p)} = E_{\text{work}} \left[ U_i \{A_j^\dagger, Y_i^\dagger; \hat{\beta}^{(p)}\} \mid \text{fitted models and recipient's observed covariates} \right],$$

where  $Y_i^\dagger$  denotes any such subsequent imputed variable; observed outcomes are kept fixed. The analysis contribution includes its original subset indicator. This expectation is used only in (3), not to replace any completed value or the analysis scores used in donor-source aggregation.

For example, if  $A$  is the response in a linear analysis with fully observed design vector  $z_i$  and fitted residual variance  $\hat{v}^{(p)}$ , then

$$U_i^{(p)} = \frac{z_i \{A_i^{(p)} - z_i^\top \hat{\beta}^{(p)}\}}{\hat{v}^{(p)}}, \quad G_{ij}^{(p)} = \frac{z_i \{a_j^{(p)} - z_i^\top \hat{\beta}^{(p)}\}}{\hat{v}^{(p)}}.$$

For a logistic analysis of  $D$  on  $A$  restricted to  $A > c_0$ , let  $z(a) = (1, a)^\top$  and  $\pi_\beta(a) = \text{expit}\{z(a)^\top \beta\}$ . Let  $h_i^{(p)}(a)$  be the probability of  $D_i = 1$  under its recorded logistic imputation model, evaluated at  $A = a$ ; if  $D_i$  is observed, set  $h_i^{(p)}(a) = D_i$ . The required expectation is

$$G_{ij}^{(p)} = \int_{c_0}^{\infty} z(a) \{h_i^{(p)}(a) - \pi_{\hat{\beta}^{(p)}}(a)\} \frac{1}{\sigma^{(p)}} \phi\left(\frac{a - a_j^{(p)}}{\sigma^{(p)}}\right) da. \quad (4)$$

The integral is not divided by the probability of inclusion: excluded records contribute zero to the estimating equation normalized by the full sample size  $n$ . It is evaluated numerically using 24-point Gauss–Legendre quadrature, truncating the standardized normal variable to  $[-10, 10]$ . Thus the definition of  $G_{ij}$  depends on the analysis model and any subsequent imputation model, while the donor probability and its derivative have the same form in both examples.

### 3.2.3 Donor-source analysis scores

Donor reuse introduces a shared observed value into several completed records. When recipient  $i_1$  and recipient  $i_2$  both copy their imputed value from the same observed donor  $j$ , the two completed values are the same observed record  $A_j^{\text{obs}}$ . Any random variation in  $A_j^{\text{obs}}$  simultaneously affects the imputed value for  $i_1$  and  $i_2$ . Their analysis scores  $U_{i_1}^{(p)}$  and  $U_{i_2}^{(p)}$  share the stochastic source  $A_j^{\text{obs}}$  across repeated samples. This is distinct from conditional independence of donor draws given a fixed observed dataset. To represent donor reuse in the variance calculation, each

recipient's analysis score is assigned to its recorded donor. Let  $U_i^{(p)}(\beta)$  be the completed-data analysis estimating function contribution for subject  $i$  in imputation  $p$ . If recipient  $i$  copied  $A$  from observed donor  $J_i^{(p)} = j$ , then the stochastic source of the copied  $A$  value is the donor record  $j$ . The donor-source analysis score for observed donor  $j$  is

$$\tilde{U}_j^{(p)}(\beta) = U_j^{(p)}(\beta) + \sum_{i: J_i^{(p)}=j} U_i^{(p)}(\beta).$$

The recipient contribution  $U_i^{(p)}(\beta)$  is evaluated using recipient  $i$ 's completed analysis record; only its stochastic source is assigned to donor  $j$  for covariance estimation. Rows whose PMM contribution has been moved to their donor source are set to zero in the donor-source representation, and rows not involved in the PMM donor map remain singleton rows. This operation preserves the estimating equation,

$$\sum_{i=1}^n U_i^{(p)}(\beta) = \sum_{i=1}^n \tilde{U}_i^{(p)}(\beta),$$

so the point estimator is unchanged; only the covariance representation of the first-order analysis scores changes. This donor-source step is the estimating-equation analogue of the hot-deck and nearest-neighbor variance idea that an observed donor is the sampling source for both its own completed record and the recipient records that copy from it (Rao and Shao, 1992; Skinner, 2002; Kim and Fuller, 2004; Yang and Kim, 2020). If donor  $j$  supplies values for recipients  $i_1, \dots, i_r$ , the ordinary rowwise meat contains

$$U_j U_j^\top + \sum_{\ell=1}^r U_{i_\ell} U_{i_\ell}^\top,$$

whereas the donor-source meat contains

$$\left( U_j + \sum_{\ell=1}^r U_{i_\ell} \right) \left( U_j + \sum_{\ell=1}^r U_{i_\ell} \right)^\top.$$

The second expression includes all cross-products among records sharing the same observed donor source. Expanding, the donor-source meat for donor  $j$  equals the rowwise meat plus the correction terms  $U_j \sum_{\ell} U_{i_\ell}^\top + \sum_{\ell} U_{i_\ell} U_j^\top + \sum_{\ell \neq \ell'} U_{i_\ell} U_{i_{\ell'}}^\top$ : the first two terms represent donor-recipient cross-products, and the third represents cross-products among recipients that share the same donor. When no recipients copy from donor  $j$  (i.e., donor  $j$  is only a self-source), the donor-source meat coincides with the rowwise meat for that record. The terms  $U_{i_\ell}$  are still the recipient-specific analysis scores computed from recipient  $i_\ell$ 's own covariates and eligibility status; the aggregation changes the source of covariance, not the completed-data estimating equation or any individual record's contribution to the point estimate. It does not assert that scores assigned to different donors are independent or that the PMM matching uncertainty is fully described by this aggregation alone.

### 3.2.4 RW components for use with PMM

For imputation  $p$ , let  $U^{(p)}$  be the  $n \times q$  matrix of rowwise analysis scores evaluated at the completed-data estimate  $\hat{\beta}^{(p)}$ , and  $\tilde{U}^{(p)}$  the corresponding donor-source matrix. The pooled

point estimate remains  $\hat{\beta} = m^{-1} \sum_p \hat{\beta}^{(p)}$ . Define

$$\tilde{U} = \frac{1}{m} \sum_{p=1}^m \tilde{U}^{(p)}, \quad \hat{\Omega}_{\text{src}} = P_n \{ \tilde{U}_i \tilde{U}_i^\top \},$$

where  $P_n f_i = n^{-1} \sum_i f_i$ . Aggregation is performed using each imputation's actual donor IDs before averaging over imputations. The PMM columns of  $\hat{\kappa}$  are given by (3). For any additional normal or logistic imputation models, their columns retain the RW-MICE calculation

$$\hat{\kappa}_{\text{par}} = \frac{1}{nm} \sum_{p=1}^m \sum_{i=1}^n U_i^{(p)} S_{\text{par},i}^{(p)\top}.$$

These columns and the PMM columns are placed in the same parameter order as the stacked influence contributions  $d_i^{(p)}$ . For the PMM regression, the observed-score linearization used in the implementation is

$$H^{(p)} = -\frac{1}{n(\sigma^{(p)})^2} \sum_{j \in \mathcal{O}_A} x_j x_j^\top, \quad d_{\text{PMM},i}^{(p)} = \begin{cases} -\{H^{(p)}\}^{-1} \frac{x_i(A_i - a_i^{(p)})}{(\sigma^{(p)})^2}, & i \in \mathcal{O}_A, \\ 0, & i \in \mathcal{M}_A. \end{cases}$$

This contribution describes the fitted matching coefficients, not the copied donor value. The PMM block has no separate residual-variance coordinate. Influence contributions for additional normal or logistic models are calculated as in Section 3.1. With the resulting stacked vector, define

$$\hat{\alpha} = \frac{1}{nm} \sum_{p=1}^m \sum_{i=1}^n d_i^{(p)} d_i^{(p)\top}, \quad \bar{d}_i = \frac{1}{m} \sum_{p=1}^m d_i^{(p)}.$$

The donor-source representation is used in  $\hat{\Omega}_{\text{src}}$  and the following cross term, but not in place of the recipient scores used to calculate  $\hat{\kappa}$ :

$$\hat{C}_{\text{src}} = \hat{\kappa} P_n \{ \bar{d}_i \tilde{U}_i^\top \} + P_n \{ \tilde{U}_i \bar{d}_i^\top \} \hat{\kappa}^\top.$$

The final RW variance estimator for use with PMM is

$$\hat{\Delta}_{\text{PMM}} = \hat{\Omega}_{\text{src}} + \hat{\kappa} \hat{\alpha} \hat{\kappa}^\top + \hat{C}_{\text{src}}, \quad \hat{\Gamma}_{\text{PMM}} = \frac{1}{n} \hat{\tau}^{-1} \hat{\Delta}_{\text{PMM}} (\hat{\tau}^{-1})^\top,$$

where

$$\hat{\tau} = \frac{1}{nm} \sum_{p=1}^m \sum_{i=1}^n \left. \frac{\partial U_i^{(p)}(\beta)}{\partial \beta^\top} \right|_{\beta=\hat{\beta}^{(p)}}.$$

This construction retains the three-term RW variance form of Section 2.4, while using a working PMM cross-moment and a donor-source analysis-score representation. Differentiability of the working probabilities does not by itself establish consistency for the actual `mice` PMM procedure. Its accuracy also depends on the matching approximation and on the working models used for  $G_{ij}$ ; these require assessment for the analysis and imputation settings under consideration.



## 4 Software Implementation

The variance estimators developed in Sections 3.1 and 3.2 require objects beyond what the standard output of the `mice` package (van Buuren and Groothuis-Oudshoorn, 2011) provides. Specifically, the imputation workflow must expose the per-imputation parameter draws  $\psi_r^{(p)}$  for each MICE block, the row-level analysis scores  $U_i^{(p)}$  from each completed-data analysis, and for PMM-imputed blocks, the donor index  $J_i^{(p)}$  mapping each recipient to its observed donor. Algorithm 1 gives the complete computational workflow; the key constraint is that  $J_i^{(p)}$  and  $\psi_r^{(p)}$  must be recorded during the imputation phase and cannot be recovered from completed datasets after the fact.

---

### Algorithm 1: RW variance estimation for MICE with optional PMM blocks

---

**Input:** Observed data  $\{O_i\}_{i=1}^n$ ; conditional imputation models for blocks  $r = 1, \dots, R$ ; number of imputations  $m$ ; analysis estimating function  $U$ ; donor pool size  $k$  for any PMM-imputed block.

**Output:** RW variance estimate  $\hat{\Gamma}$ .

---

**Step 1 (Impute).**

**for**  $p = 1, \dots, m$  **do**

For each parametric block  $r$ : draw  $\psi_r^{(p)}$  from its large-sample posterior and impute the block’s missing values; **record**  $\psi_r^{(p)}$ .

For each PMM block: draw  $\psi^{(p)}$ , form predictive means  $\eta_i(\psi^{(p)})$ , calibrate the bandwidth  $\lambda_i$  by the empirical rule to target  $k$  effective donors, draw donor  $J_i^{(p)}$  from the smooth donor law  $p_{ij}(\psi^{(p)})$ , and copy  $A_i^{(p)} = A_{J_i^{(p)}}^{\text{obs}}$ ; **record**  $\psi^{(p)}$  and  $J_i^{(p)}$ .

**end for**

**Step 2 (Analyze).** For each completed dataset, solve the complete-data estimating equation; average the solutions to obtain  $\hat{\beta}$ . Evaluate the row-level scores  $U_i^{(p)}(\hat{\beta})$  and the bread  $\hat{\tau}$ .

**Step 3 (Scores).** For each block  $r$ , evaluate  $S_{r,\text{mis},i}^{(p)}$  and  $d_{r,i}^{(p)}$  at the recorded  $\psi_r^{(p)}$  (Section 3.1); for a PMM block, evaluate  $S_{\text{PMM},i}^{(p)}$  (Section 3.2.1) and aggregate analysis scores to the donor source,  $\tilde{U}^{(p)}$ , using the recorded  $J_i^{(p)}$  (Section 3.2.3). Stack across blocks.

**Step 4 (Assemble).** Form  $\hat{\Omega}$  (or  $\hat{\Omega}_{\text{src}}$ ),  $\hat{\kappa}$ ,  $\hat{\alpha}$ ,  $\hat{C}$  (or  $\hat{C}_{\text{src}}$ ), set  $\hat{\Delta} = \hat{\Omega} + \hat{\kappa}\hat{\alpha}\hat{\kappa}^\top + \hat{C}$ .

**return**  $\hat{\Gamma} = n^{-1}\hat{\tau}^{-1}\hat{\Delta}(\hat{\tau}^{-1})^\top$ .

---

We have developed an R package that is designed to intake a `mice` imputation object and return RW variance estimators using similar syntax to the `pool` function in `mice` that returns RR variance estimators. The package is available at <https://github.com/LucyMcGowan/rw>.

## 5 Experiments with simulated data

### 5.1 Estimators and evaluation metrics

We compare four variance estimators in two imputation families. For chained parametric MICE, RW-MICE denotes our stacked Robins–Wang estimator (Section 3.1) and RR-MICE the Rubin’s-rules variance applied to the same imputations. For predictive mean matching,

RW-PMM denotes the Robins–Wang estimator for use with PMM (Section 3.2) and RR-PMM the Rubin’s-rules variance applied to the PMM imputations. Within each RW/RR pair the point estimator  $\hat{\beta}$  is identical; only the variance estimator differs. The two simulation settings are chosen to stress-test the two extensions developed in Sections 3.1 and 3.2 in qualitatively different ways. The first setting, the missing-data benchmark of Robins and Wang (2000), uses ordinary outcome missingness and a linear analysis estimating equation. Because the data-generating mechanism is well understood, we know in advance which direction RR should fail: anti-conservative when the analysis exploits heteroskedastic structure that the imputation model does not reflect (panel (b) of Figure 1), and conservative when the imputation model is richer than the analysis (panel (c)). This allows the RW-MICE extension to be evaluated against a known calibration benchmark rather than against empirical performance alone. The benchmark also includes PMM variants, which assess whether the smooth copied-donor correction introduces any artefact under ordinary missingness. The second setting, the EHR validation setting of Giganti and Shepherd (2020), is a more demanding test that combines both extensions simultaneously. Two variables are imputed by MICE, one continuous and one binary. The continuous variable is imputed by either a Gaussian parametric model or by PMM; the binary variable is imputed by a logistic conditional model. Critically, the imputed continuous variable determines both the analysis covariate and the eligibility criterion for the analysis subset. This imputation structure is the canonical case of conservative uncongeniality: the imputation model is fitted on all subjects, but the analysis equation is evaluated in a subset whose membership fluctuates with the imputed values, inflating the between-imputation variance for Rubin’s rules. For each simulation scenario,  $\beta^*$  denotes the large-sample target of the implemented imputation-and-analysis procedure; in finite simulation summaries it is approximated by the Monte Carlo mean of the multiply imputed analysis estimator. The primary variance calibration measure is coverage of  $\beta^*$ : for each replicate, the usual confidence interval  $\hat{\beta} \pm 1.96\widehat{SE}$  is checked for coverage of  $\beta^*$ . This isolates variance calibration from finite-sample bias. We also report coverage of the data-generating coefficient  $\beta_0$ , mean bias  $\hat{\beta} - \beta_0$ , the empirical standard deviation of the Monte Carlo estimates, and the mean estimated standard error. Thus, coverage of  $\beta^*$  answers whether the standard error estimates the sampling variability of the implemented estimator, whereas coverage of  $\beta_0$  also reflects finite-sample bias and target mismatch. All simulations use 2500 Monte Carlo replications per cell.

## 5.2 Missing-data benchmark

The missing-data benchmark follows the original simulation structure used by RW: a hypothetical example involving children at a daycare center. Each subject has a group indicator  $A$ , an age covariate  $X$ , and a continuous outcome  $Z$ . We use  $A = 0$  for infants and  $A = 1$  for toddlers, with  $P(A = 1) = 1/3$ . Infants have  $X \sim U(0.1, 0.8)$  and toddlers have  $X \sim U(0.8, 2.0)$ . In the main variants,

$$Z = X + \varepsilon, \quad \varepsilon \mid X, A \sim N\{0, X^{\eta_A}\}.$$

The outcome is fully observed for infants. Among toddlers,  $Z$  is missing with probability 0.60 except in the outcome-dependent observation variant, where  $V = 1$  denotes that  $Z$  is observed. The completed-data analysis is the no-intercept least-squares estimating equation for the coefficient of  $X$  in  $E(Z \mid X) = \beta X$ . In one variant the analysis is restricted to toddlers; otherwise all subjects are included. We generated data with both  $n = 150$ , matching the

original RW simulations exactly, and  $n = 1000$ . The six scenarios are:

- S1:  $\eta_0 = \eta_1 = 0$ , analysis restricted to toddlers;
- S2a:  $\eta_0 = \eta_1 = 1$ , all-subject analysis;
- S2b:  $\eta_0 = \eta_1 = -1$ , all-subject analysis;
- S2c:  $\eta_0 = 2, \eta_1 = 1$ , all-subject analysis;
- S3a:  $\eta_0 = \eta_1 = 0, P(V = 1 | Z, A = 1) = \text{expit}(2 - 0.5Z)$ ;
- S3b:  $Z = X + 0.5X^2 + \varepsilon, \eta_0 = \eta_1 = 0$ .

S1 is the subgroup analysis case. S2a–S2c retain the linear mean analysis but change the variance relationship between the imputation and analysis estimating equations. S3a changes the observation mechanism among toddlers, and S3b changes the outcome mean while retaining the linear analysis. In S3a and S3b, coverage of  $\beta_0$  can differ from coverage of  $\beta^*$  because the finite-sample estimator is centered away from the data-generating coefficient. Tables 1 and 2 summarize the missing-data benchmark by scenario. For parametric MICE, RW/RR pairs compare RW-MICE with RR-MICE. For PMM, RW/RR pairs compare RW-PMM with RR-PMM, using donor pool size  $k = 5$ . RW-MICE has coverage of  $\beta^*$  close to nominal in both sample size settings. RW-PMM shows the same pattern at  $n = 1000$ , with coverage of  $\beta^*$  between 0.945 and 0.965 across the six scenarios. The  $n = 150$  PMM cells are more variable, as expected for small samples with copied donor values, but still correct the most anti-conservative RR-PMM cells in S1, S2a, S2c, and S3b. RR can be far from nominal: anti-conservative in S2a and S2c, and conservative in S2b. These results reproduce the key RW message in a chained-equation implementation: when imputation and analysis estimating equations are uncongenial, variance calibration depends on the estimating-equation structure rather than on RR alone.

| Scenario | Imputation model | Mean estimate | Bias   | Empirical SD | Mean estimated SE RW/RR | Coverage $\beta^*$ RW/RR | Coverage $\beta_0$ RW/RR | Median corr. (score comp.) |
|----------|------------------|---------------|--------|--------------|-------------------------|--------------------------|--------------------------|----------------------------|
| S1       | Parametric MICE  | 0.999         | -0.001 | 0.109        | 0.095 / 0.101           | 0.936 / 0.948            | 0.936 / 0.948            |                            |
|          | PMM ( $k = 5$ )  | 0.981         | -0.019 | 0.174        | 0.234 / 0.124           | 0.925 / 0.832            | 0.920 / 0.832            |                            |
| S2a      | Parametric MICE  | 0.998         | -0.002 | 0.108        | 0.095 / 0.066           | 0.942 / 0.806            | 0.943 / 0.804            |                            |
|          | PMM ( $k = 5$ )  | 0.990         | -0.010 | 0.164        | 0.182 / 0.104           | 0.926 / 0.783            | 0.925 / 0.778            |                            |
| S2b      | Parametric MICE  | 0.999         | -0.001 | 0.089        | 0.079 / 0.137           | 0.947 / 0.998            | 0.947 / 0.998            |                            |
|          | PMM ( $k = 5$ )  | 0.968         | -0.032 | 0.124        | 0.308 / 0.153           | 0.970 / 0.985            | 0.972 / 0.987            |                            |
| S2c      | Parametric MICE  | 0.998         | -0.002 | 0.107        | 0.094 / 0.055           | 0.945 / 0.722            | 0.945 / 0.718            |                            |
|          | PMM ( $k = 5$ )  | 0.992         | -0.008 | 0.168        | 0.165 / 0.097           | 0.913 / 0.736            | 0.916 / 0.731            |                            |
| S3a      | Parametric MICE  | 0.938         | -0.062 | 0.070        | 0.061 / 0.063           | 0.943 / 0.955            | 0.735 / 0.755            |                            |
|          | PMM ( $k = 5$ )  | 0.939         | -0.061 | 0.098        | 0.106 / 0.095           | 0.945 / 0.941            | 0.909 / 0.890            |                            |
| S3b      | Parametric MICE  | 1.670         | 0.670  | 0.097        | 0.085 / 0.086           | 0.944 / 0.941            | 0.002 / 0.002            |                            |
|          | PMM ( $k = 5$ )  | 1.667         | 0.667  | 0.147        | 0.140 / 0.109           | 0.931 / 0.858            | 0.012 / 0.002            |                            |

Table 1: Missing-data benchmark at  $n = 150$  and  $m = 20$ , with 2500 Monte Carlo replications per scenario. Bias is the Monte Carlo mean estimate minus the data-generating coefficient. RW/RR pairs refer to RW-MICE/RR-MICE for parametric MICE rows and RW-PMM/RR-PMM for PMM rows.

These results confirm that the MICE stacking construction of Section 3.1 correctly captures the three RW variance sources under chained conditional imputation. Second, the RW-PMM estimator inherits this calibration at  $n = 1000$  and makes substantial improvements over RR-PMM even at  $n = 150$ , where small sample variance is larger. Both estimators correctly distinguish between coverage of  $\beta^*$  (a pure variance diagnostic, near-nominal for the proposed

| Scenario | Imputation model | Mean estimate | Bias   | Empirical SD | Mean estimated SE<br>RW/RR | Coverage $\beta^*$<br>RW/RR | Coverage $\beta_0$<br>RW/RR | Median corr.<br>(score comp.) |
|----------|------------------|---------------|--------|--------------|----------------------------|-----------------------------|-----------------------------|-------------------------------|
| S1       | Parametric MICE  | 1.000         | -0.000 | 0.059        | 0.059 / 0.060              | 0.950 / 0.952               | 0.950 / 0.953               |                               |
|          | PMM ( $k = 5$ )  | 0.996         | -0.004 | 0.063        | 0.065 / 0.047              | 0.948 / 0.859               | 0.949 / 0.860               |                               |
| S2a      | Parametric MICE  | 0.999         | -0.001 | 0.059        | 0.060 / 0.040              | 0.952 / 0.812               | 0.952 / 0.810               |                               |
|          | PMM ( $k = 5$ )  | 1.000         | 0.000  | 0.065        | 0.065 / 0.041              | 0.952 / 0.784               | 0.952 / 0.785               |                               |
| S2b      | Parametric MICE  | 0.999         | -0.001 | 0.045        | 0.046 / 0.082              | 0.950 / 1.000               | 0.951 / 1.000               |                               |
|          | PMM ( $k = 5$ )  | 0.996         | -0.004 | 0.046        | 0.053 / 0.055              | 0.965 / 0.982               | 0.965 / 0.982               |                               |
| S2c      | Parametric MICE  | 0.999         | -0.001 | 0.059        | 0.059 / 0.034              | 0.955 / 0.725               | 0.953 / 0.726               |                               |
|          | PMM ( $k = 5$ )  | 0.999         | -0.001 | 0.064        | 0.064 / 0.039              | 0.945 / 0.762               | 0.945 / 0.760               |                               |
| S3a      | Parametric MICE  | 0.939         | -0.061 | 0.038        | 0.038 / 0.038              | 0.949 / 0.954               | 0.626 / 0.645               |                               |
|          | PMM ( $k = 5$ )  | 0.939         | -0.061 | 0.038        | 0.039 / 0.037              | 0.954 / 0.941               | 0.658 / 0.618               |                               |
| S3b      | Parametric MICE  | 1.672         | 0.672  | 0.052        | 0.052 / 0.052              | 0.953 / 0.945               | 0.000 / 0.000               |                               |
|          | PMM ( $k = 5$ )  | 1.688         | 0.688  | 0.054        | 0.055 / 0.042              | 0.954 / 0.868               | 0.000 / 0.000               |                               |

Table 2: Missing-data benchmark at  $n = 1000$  and  $m = 50$ , with 2500 Monte Carlo replications per scenario. Columns are defined as in Table 1.

methods) and coverage of  $\beta_0$  (which also reflects the bias of the implemented estimator, and is low when  $\beta^* \neq \beta_0$  as in S3a and S3b). The benchmark setting, however, does not fully stress test the PMM correction. The analysis subset does not depend on the imputed variable: whether  $Z$  is large or small does not change which subjects enter the least-squares estimating equation. The donor-source correlation adjustment (Section 3.2.3) therefore has limited impact in this setting, because the analysis-score contributions of recipients with a common donor are not strongly correlated through a shared eligibility event. The GS validation setting below provides the harder test, where the copied donor value simultaneously appears in the analysis as a covariate and determines the analysis subset.

### 5.3 Measurement-error validation setting

The second simulation setting follows the GS electronic health record validation example (Giganti and Shepherd, 2020). The goal is to estimate the association between a continuous exposure  $A$  and a binary outcome  $D$ . Only error-prone versions  $A^*$  and  $D^*$  are available for every subject, while the validated variables  $A$  and  $D$  are observed in a validation subset; the validated values are imputed for the remaining subjects. The analysis model is a logistic regression of  $D$  on  $A$ , restricted to the post-imputation analysis set  $\{A > 2\}$ ; the eligibility rule reflects the common situation in which the study question concerns only subjects above a clinical threshold on the (error-prone, then imputed) exposure. We generated data with  $n = 2000$  and  $n = 4000$ , the latter matching the original GS implementation; the full sample-size grid  $n \in \{2000, 4000, 6000, 8000\}$  for parametric MICE is shown in Figure 5. This setting is more demanding than the RW benchmark in three concrete ways. First, there are two MICE blocks, one continuous and one binary, so the stacking must handle a mixed-model chained structure with cross-block nuisance terms as in Section 3.1. Second, the eligibility rule  $\{A > 2\}$  means that the imputed  $A$  value determines whether a subject’s analysis score is zero or nonzero: two recipients copying from the same donor therefore have correlated eligibility decisions and correlated analysis score values, making this the setting where the donor source adjustment of Section 3.2.3 matters most. Third, the imputation model is fitted on all subjects while the analysis runs only in the post imputation subset  $\{A > 2\}$ , the canonical case of conservative uncongeniality, which inflates the between-imputation variance through random subset membership, reproducing the panel (c) pattern of Figure 1. As in the previous section, we implement both parametric and PMM imputation. In the parametric MICE cells,  $A$  is imputed from a Gaussian conditional working model with predictors  $X_1, X_2, A^*, D^*$ , and  $D$  from a logistic model with the same predictors plus the completed  $A$ . In the PMM cells, only the continuous  $A$  block is replaced by the smooth copied donor law; the  $D$  block remains logistic. This design isolates the PMM correction: the binary block contributes through the standard RW-MICE framework, while the donor source adjustment applies only to the  $A$  block that directly drives eligibility. Tables 3 and 4 report results separately by sample size. Each table uses the same layout and presents three imputation models: parametric MICE, PMM with  $k = 5$ , and PMM with  $k = 10$ . We vary the number of imputations  $m \in \{5, 25, 50, 100\}$  to assess sensitivity to the number of imputations, and the donor pool size  $k$  because it governs how widely the smooth law spreads probability over donors; Results for the larger donor pools  $k = 20$  and  $k = 30$  are reported in Appendix B (Table 5; Figure 3) and are broadly similar to those for  $k = 5$  and  $k = 10$ ; results across sample sizes for  $k = 5$  are shown in Figure 4.

| $m$ | Imputation model | Mean estimate | Bias   | Empirical SD | Mean estimated SE RW/RR | Coverage $\beta^*$ RW/RR | Coverage $\beta_0$ RW/RR | Median corr. (score comp.) |
|-----|------------------|---------------|--------|--------------|-------------------------|--------------------------|--------------------------|----------------------------|
| 5   | Parametric MICE  | 0.850         | -0.040 | 0.120        | 0.114 / 0.166           | 0.937 / 0.985            | 0.900 / 0.973            |                            |
| 5   | PMM ( $k = 5$ )  | 0.851         | -0.041 | 0.134        | 0.177 / 0.168           | 0.979 / 0.976            | 0.968 / 0.966            |                            |
| 5   | PMM ( $k = 10$ ) | 0.842         | -0.046 | 0.136        | 0.149 / 0.169           | 0.956 / 0.978            | 0.944 / 0.962            |                            |
| 25  | Parametric MICE  | 0.850         | -0.040 | 0.109        | 0.107 / 0.162           | 0.944 / 0.996            | 0.910 / 0.985            |                            |
| 25  | PMM ( $k = 5$ )  | 0.854         | -0.037 | 0.125        | 0.136 / 0.165           | 0.963 / 0.991            | 0.947 / 0.978            |                            |
| 25  | PMM ( $k = 10$ ) | 0.843         | -0.048 | 0.127        | 0.129 / 0.166           | 0.956 / 0.989            | 0.923 / 0.973            |                            |
| 50  | Parametric MICE  | 0.850         | -0.040 | 0.108        | 0.107 / 0.162           | 0.946 / 0.998            | 0.907 / 0.984            |                            |
| 50  | PMM ( $k = 5$ )  | 0.855         | -0.038 | 0.124        | 0.130 / 0.164           | 0.957 / 0.992            | 0.934 / 0.977            |                            |
| 50  | PMM ( $k = 10$ ) | 0.843         | -0.048 | 0.125        | 0.126 / 0.165           | 0.946 / 0.989            | 0.929 / 0.978            |                            |
| 100 | Parametric MICE  | 0.850         | -0.040 | 0.107        | 0.106 / 0.162           | 0.944 / 0.998            | 0.909 / 0.986            |                            |
| 100 | PMM ( $k = 5$ )  | 0.849         | -0.040 | 0.123        | 0.127 / 0.164           | 0.954 / 0.992            | 0.940 / 0.979            |                            |
| 100 | PMM ( $k = 10$ ) | 0.843         | -0.048 | 0.124        | 0.124 / 0.165           | 0.953 / 0.991            | 0.934 / 0.986            |                            |

Table 3: Measurement-error validation setting at  $n = 2000$ , with 2500 Monte Carlo replications per completed cell. RW/RR pairs refer to RW-MICE/RR-MICE for parametric MICE rows and RW-PMM/RR-PMM for PMM rows.

| $m$ | Imputation model | Mean estimate | Bias   | Empirical SD | Mean estimated SE RW/RR | Coverage $\beta^*$ RW/RR | Coverage $\beta_0$ RW/RR | Median corr. (score comp.) |
|-----|------------------|---------------|--------|--------------|-------------------------|--------------------------|--------------------------|----------------------------|
| 5   | Parametric MICE  | 0.869         | -0.020 | 0.083        | 0.082 / 0.117           | 0.944 / 0.989            | 0.921 / 0.980            |                            |
| 5   | PMM ( $k = 5$ )  |               |        |              |                         |                          |                          |                            |
| 5   | PMM ( $k = 10$ ) |               |        |              |                         |                          |                          |                            |
| 25  | Parametric MICE  | 0.869         | -0.021 | 0.076        | 0.078 / 0.115           | 0.951 / 0.996            | 0.920 / 0.992            |                            |
| 25  | PMM ( $k = 5$ )  |               |        |              |                         |                          |                          |                            |
| 25  | PMM ( $k = 10$ ) |               |        |              |                         |                          |                          |                            |
| 50  | Parametric MICE  | 0.869         | -0.021 | 0.075        | 0.077 / 0.115           | 0.950 / 0.995            | 0.920 / 0.992            |                            |
| 50  | PMM ( $k = 5$ )  |               |        |              |                         |                          |                          |                            |
| 50  | PMM ( $k = 10$ ) |               |        |              |                         |                          |                          |                            |
| 100 | Parametric MICE  | 0.869         | -0.021 | 0.075        | 0.077 / 0.114           | 0.956 / 0.997            | 0.925 / 0.992            |                            |
| 100 | PMM ( $k = 5$ )  |               |        |              |                         |                          |                          |                            |
| 100 | PMM ( $k = 10$ ) |               |        |              |                         |                          |                          |                            |

Table 4: Measurement-error validation setting at  $n = 4000$ , with 2500 Monte Carlo replications per completed cell. Columns are defined as in Table 3.

Tables 3 and 4 show that RW-MICE tracks the empirical standard deviation more closely than RR in the parametric MICE rows across all values of  $m$ , confirming that the MICE extension handles the conservative uncongeniality of this setting. RR consistently overstates the standard error by approximately 30–50%, reproducing the GS finding at these larger sample sizes (Figure 2). For PMM, the empirical rule concentrates the smooth law’s probability on approximately  $k$  donors while assigning positive probability to the remainder; larger  $k$  spreads probability over more distant donors. For the completed  $n = 2000$  PMM rows, RW-PMM standard errors are close to the empirical standard deviation for  $m \geq 25$ , while RR is conservative for the same PMM point estimator. Across the reported  $n = 2000$  PMM validation cells with  $k \in \{5, 10, 20, 30\}$ , coverage of  $\beta^*$  for RW-PMM ranges from 0.943 to 0.979, compared with 0.976 to 0.992 for RR-PMM. Coverage of  $\beta_0$  is more sensitive to  $k$  because larger donor pools admit more distant donors (a wider *donor width*, i.e. the spread of predictive means among eligible donors), which shifts the PMM point estimator away from  $\beta_0$ ; this reflects donor-width bias rather than variance miscalibration. Results for the larger donor pools  $k = 20$  and  $k = 30$  are reported in Appendix B and are broadly similar to those for  $k = 5$  and  $k = 10$ .

## 5.4 Summary

Across both settings the proposed RW estimators achieve near-nominal coverage of  $\beta^*$  in most cases, while RR fails in predictable directions: anti-conservative when the analysis exploits structure the imputation model does not reflect, and conservative when the imputation model uses information beyond the analysis. These failures are not small: they reach 14–23 coverage-point shortfalls in the anti-conservative direction (RW benchmark, S2a, S2c, S3b) and approximately 30–50% SE inflation in the conservative direction (GS validation setting). The RW-PMM estimator inherits the calibration of RW-MICE in the benchmark setting and correctly handles the eligibility-by-imputation correlation structure in the GS setting, with coverage of  $\beta^*$  between 0.943 and 0.979 across the completed PMM cells. The distinction between coverage of  $\beta^*$  and coverage of  $\beta_0$ , illustrated in Figure 1 and maintained throughout the simulation tables, is important: when the implemented estimator is biased for  $\beta_0$  (as in S3a, S3b, and the GS validation setting), coverage of  $\beta^*$  is the correct measure of variance calibration, and the proposed RW methods achieve it.

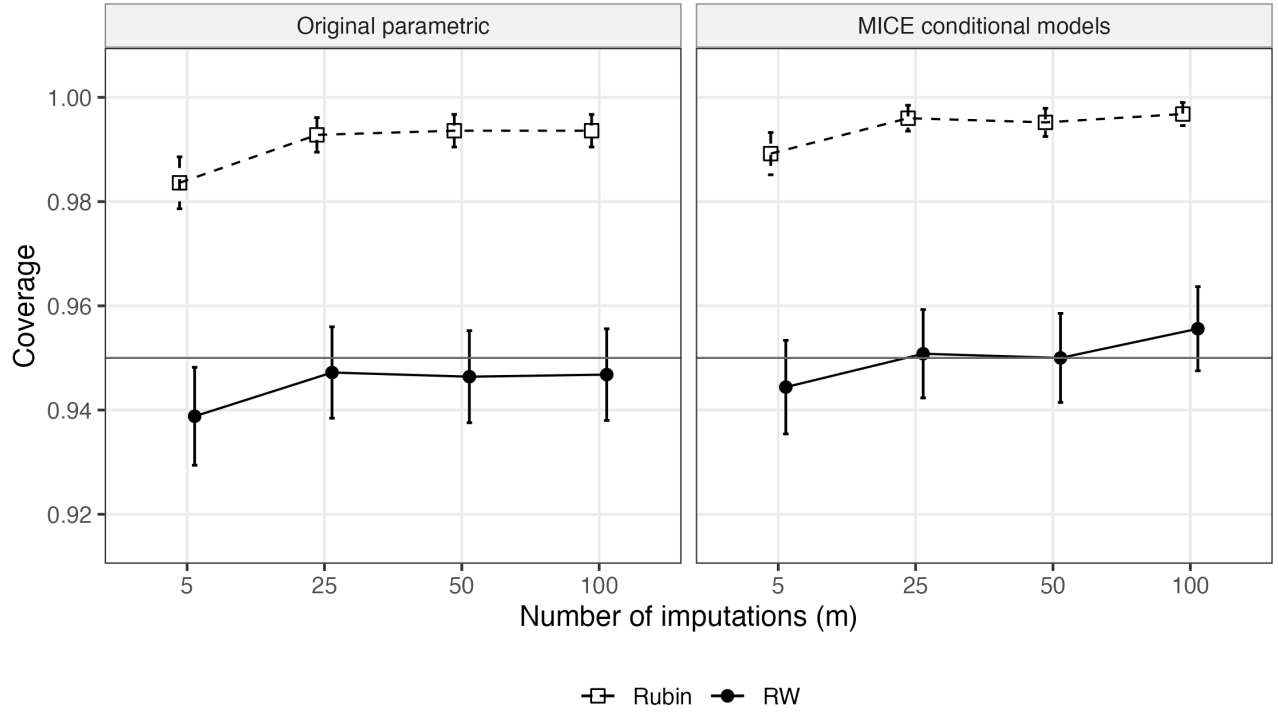


Figure 2: Empirical-centered coverage in the Giganti–Shepherd measurement-error setting with parametric imputation at  $n = 4000$ . The left panel shows the original Giganti–Shepherd parametric implementation, and the right panel shows the chained conditional MICE implementation.



## 6 Discussion

This paper develops two extensions of the estimating-equation variance framework of Robins and Wang (2000) to settings that arise routinely in modern medical data analysis: chained conditional (MICE) imputation and predictive mean matching. The extensions are not ad hoc adjustments but follow directly from the RW construction once the relevant score and nuisance objects are identified for each setting. Both extensions are validated in simulation and the key implementation objects are exposed by a companion R package, `rw`, that seamlessly integrates with `mice` package output. Our open-source software is easy to use because it mirrors the functions already in place for Rubin’s rules variance estimators and takes as input a `mice` package imputed dataset. The failure of Rubin’s rules under uncongeniality has been understood since Meng (1994) and formalised for estimating equation analyses by Wang and Robins (1998). The present work bridges these theoretical results and applied practice by showing that both failure modes (anti-conservative and conservative) arise in common MICE and PMM workflows and can be corrected within the same sandwich framework. The GS validation setting, where eligibility is determined by the imputed variable and Rubin’s rules overstate the variance by approximately 30–50%, is an instance of Meng’s conservative uncongeniality that has become increasingly common in EHR based analyses with imputed inclusion criteria (Shen et al., 2026). The RW-MICE and RW-PMM estimators restore calibration in this setting without requiring any change to the imputation or analysis workflow.

**The bootstrap.** As noted in Section 1, the bootstrap is an alternative to analytic variance estimation under uncongeniality: re-running the full imputation and analysis procedure across bootstrap samples consistently estimates the variance of the implemented estimator regardless of congeniality (Schomaker and Heumann, 2018; Bartlett and Hughes, 2020). The RW estimator has a practical advantage when the imputation workflow is expensive to repeat.

**Limitations.** Several limitations should be noted. First, the PMM extension requires donor indices  $J_i^{(p)}$  to be recorded during imputation; they cannot be reconstructed from completed datasets. This is a software requirement rather than a theoretical obstacle, and the companion package handles it automatically, but practitioners using standard `mice` output will need to update their imputation workflow. Second, the present work considers the case where one continuous variable is imputed by the smooth copied donor law and the remaining blocks are parametric. When several variables are imputed by PMM with potentially different donor maps, the multi-source covariance structure requires additional work. The analysis score contributions of two recipients can be correlated through two independent donor maps simultaneously, and stacking the donor source representations from different PMM blocks does not automatically account for this joint dependence. Possible extensions include block PMM with common donors or a multi-way source aggregation; these fit within the same source-aggregation framework, and we plan to support general PMM problems in a future release of the package. Third, the smooth copied donor law is a working device for computing the RW score; it does not change the imputed values. The asymptotic equivalence between the RW estimator for the smooth law and the bootstrap variance of the hard nearest-neighbor PMM estimator has not been formally established. The simulation evidence is supportive, but a rigorous analysis of the approximation error as  $k \rightarrow \infty$  or  $\lambda \rightarrow \infty$  remains an open problem. Fourth, the paper focuses on Gaussian and logistic analysis estimating equations. Extensions to survival analysis, competing risks, or

semiparametric models with partially observed outcomes are not considered, though the RW framework is in principle applicable to any regular estimating equation.

## Acknowledgments

This manuscript was supported in part by the U.S. National Institutes of Health (NIH) grant R37 AI131771 (BDW) and XX (LDM).

## Code availability

We provide code to reproduce both simulations and data analyses on GitHub at `replacewithURL`.

## A Consistency Arguments

### A.1 Chained conditional imputation models

Let  $p = 1, \dots, m$  index the completed datasets, with  $m$  fixed. For imputation  $p$ , write the completed data analysis contribution as  $U_i^{(p)}(\beta)$  and define  $\bar{U}_i(\beta) = m^{-1} \sum_{p=1}^m U_i^{(p)}(\beta)$ . The RW-MICE estimator solves  $0 = P_n \bar{U}(\hat{\beta})$ .

#### Conditions.

- (i)  $E\{\bar{U}_i(\beta)\}$  has a unique zero  $\beta^*$  and  $\tau = E\{\partial \bar{U}_i(\beta^*)/\partial \beta^\top\}$  is nonsingular.
- (ii) A law of large numbers and central limit theorem apply to  $\bar{U}_i(\beta^*)$  and to its derivative.
- (iii) For each block  $r$ , the realized conditional model state satisfies  $\psi_r^{(p)} - \psi_{r0} = O_p(n^{-1/2})$ , where  $\psi_{r0}$  is the probability limit of the block  $r$  parameter estimator under repeated sampling.
- (iv) For each block  $r$ , the matrix  $A_r^{(p)} = P_n\{\partial s_{r,i}^{(p)}(\psi_r)/\partial \psi_r^\top|_{\psi_r=\psi_r^{(p)}} \delta_{r,i}\}$  converges in probability to a nonsingular limit, and the block scores have finite second moments.

**Result.** Under conditions (i)–(iv),

$$\sqrt{n}(\hat{\beta} - \beta^*) \xrightarrow{d} N\{0, \tau^{-1}(\Omega_m + \kappa_m \alpha_m \kappa_m^\top + C_m)(\tau^{-1})^\top\},$$

where the population moments  $\Omega_m, \kappa_m, \alpha_m, C_m$  are defined below, and the estimator  $\hat{\Gamma}_{\text{RW-MICE}} = n^{-1} \hat{\tau}^{-1}(\hat{\Omega}_m + \hat{\kappa}_m \hat{\alpha}_m \hat{\kappa}_m^\top + \hat{C}_m)(\hat{\tau}^{-1})^\top$  is consistent.

**Argument.** For block  $r$ , define the missing value score  $S_{r,\text{mis},i}^{(p)} = s_{r,i}^{(p)}(\psi_r^{(p)})$  for rows imputed in block  $r$  and zero otherwise, and the observed side score  $S_{r,\text{obs},i}^{(p)} = \delta_{r,i} s_{r,i}^{(p)}(\psi_r^{(p)})$ . The blockwise nuisance influence is  $d_{r,i}^{(p)} = -\{A_r^{(p)}\}^{-1} S_{r,\text{obs},i}^{(p)}$ . By condition (iii),  $\psi_r^{(p)} - \psi_{r0} = O_p(n^{-1/2})$ . A first-order Taylor expansion of any smooth function of  $\psi_r^{(p)}$  around  $\psi_{r0}$  therefore changes the empirical RW moments only at order  $n^{-1/2}$ , which is higher order than the leading variance term. The substitution error from using realized states  $\psi_r^{(p)}$  as plug-ins is asymptotically negligible. Stacking across blocks gives the composite missing value score  $S_{\text{mis},i}^{(p)} = (S_{1,\text{mis},i}^{(p)}, \dots, S_{R,\text{mis},i}^{(p)})^\top$  and nuisance influence  $d_i^{(p)} = (d_{1,i}^{(p)}, \dots, d_{R,i}^{(p)})^\top$ . No joint likelihood across blocks is required; each block contributes its own blockwise Newton linearization. Define the fixed- $m$  population moments

$$\begin{aligned}\Omega_m &= E\{\bar{U}_i(\beta^*)\bar{U}_i(\beta^*)^\top\}, \\ \kappa_m &= m^{-1} \sum_{p=1}^m E\{U_i^{(p)}(\beta^*) S_{\text{mis},i}^{(p)\top}\}, \quad \alpha_m = m^{-1} \sum_{p=1}^m E\{d_i^{(p)} d_i^{(p)\top}\}, \\ C_m &= \kappa_m E(\bar{d}_i \bar{U}_i^\top) + E(\bar{U}_i \bar{d}_i^\top) \kappa_m^\top, \quad \bar{d}_i = m^{-1} \sum_{p=1}^m d_i^{(p)}.\end{aligned}$$

By conditions (ii) and (iv) and the plug-in argument,  $\hat{\Omega}_m \xrightarrow{p} \Omega_m$ ,  $\hat{\kappa}_m \xrightarrow{p} \kappa_m$ ,  $\hat{\alpha}_m \xrightarrow{p} \alpha_m$ ,  $\hat{C}_m \xrightarrow{p} C_m$ . A Taylor expansion of  $P_n \bar{U}(\hat{\beta})$  around  $\beta^*$  gives  $\sqrt{n}(\hat{\beta} - \beta^*) = -\tau^{-1} \sqrt{n} P_n \bar{U}(\beta^*) + o_p(1)$ . The variance of  $\sqrt{n} P_n \bar{U}(\beta^*)$  decomposes into  $\Omega_m + \kappa_m \alpha_m \kappa_m^\top + C_m$  by the standard RW argument, with the stacked  $S_{\text{mis}}$  and  $d$  vectors in place of the single model quantities. Substituting the consistent empirical moments and applying the continuous mapping theorem and Slutsky's theorem yields the stated result. When  $R = 1$  the stacked score reduces to the single model score and the estimator coincides with the original RW estimator of Robins and Wang (2000).

## A.2 Bridging the two settings

The MICE argument above covers the case where every block is parametric. The most common applied setting additionally contains a PMM block: one continuous variable is imputed by the smooth copied donor law and the remaining blocks are parametric. In this case, the argument of Section A.1 applies to the parametric blocks, the PMM block contributes its smooth donor score  $S_{\text{PMM},i}^{(p)}$  and predictive mean nuisance influence  $d_i^{(p)}$  to the stacked vectors, and the combined quantities enter the PMM argument of Section A.3 below, which additionally corrects the analysis score covariance for the shared donor dependence induced by the copied value feature.

## A.3 Smooth copied donor PMM

The proposed PMM estimator uses a smooth copied donor working law: the completed value is still copied from an observed donor, but the donor probability is a differentiable function of the predictive mean distance. The argument below concerns this smooth law and is not a claim that the nondifferentiable hard nearest-neighbor donor rule has the same nuisance score. Let  $J_i^{(p)}$  be the selected donor for recipient  $i$  in imputation  $p$ , and let  $S_{\text{PMM},i}^{(p)} = \partial \log p_{i,J_i^{(p)}}(\psi) / \partial \psi|_{\psi=\psi^{(p)}}$  be the score of the smooth donor law, with zero contribution for rows not imputed by the PMM block.

**Additional conditions.** In addition to conditions (i)–(iv) applied to the parametric blocks, assume that the smooth donor law is differentiable in  $\psi$  with finite second moments for  $S_{\text{PMM},i}$ ,  $d_i$ , and  $U_i$ , and that the donor source analysis score array satisfies a Lindeberg condition: no single donor source dominates the averaged meat, in the sense that  $n^{-1} \max_i \|\tilde{U}_i\|^2 \xrightarrow{p} 0$  and  $P_n(\tilde{U}_i \tilde{U}_i^\top)$  converges.

**Donor source scores.** The donor source analysis score for donor  $j$  is

$$\tilde{U}_j^{(p)}(\beta) = U_j^{(p)}(\beta) + \sum_{i: J_i^{(p)}=j} U_i^{(p)}(\beta).$$

Define  $\tilde{U}_i(\beta) = m^{-1} \sum_{p=1}^m \tilde{U}_i^{(p)}(\beta)$ ,  $\bar{d}_i = m^{-1} \sum_{p=1}^m d_i^{(p)}$ , and the population moments

$$\begin{aligned} \Omega_{\text{src}} &= E\{\tilde{U}_i(\beta^*) \tilde{U}_i(\beta^*)^\top\}, \quad \kappa = m^{-1} \sum_{p=1}^m E\{U_i^{(p)}(\beta^*) S_i^{(p)\top}\}, \\ \alpha &= m^{-1} \sum_{p=1}^m E\{d_i^{(p)} d_i^{(p)\top}\}, \quad C_{\text{src}} = \kappa E(\bar{d}_i \tilde{U}_i^\top) + E(\tilde{U}_i \bar{d}_i^\top) \kappa^\top. \end{aligned}$$

**Result.** Under the above conditions,

$$\sqrt{n}(\hat{\beta} - \beta^*) \xrightarrow{d} N\{0, \tau^{-1}(\Omega_{\text{src}} + \kappa \alpha \kappa^\top + C_{\text{src}})(\tau^{-1})^\top\},$$

and the estimator  $\hat{\Gamma}_{\text{PMM}}$  obtained by replacing population moments with their sample analogues is consistent.

**Argument.** The donor source representation preserves the averaged estimating equation:  $\sum_i U_i^{(p)} = \sum_i \tilde{U}_i^{(p)}$  for each  $p$ , so the bread matrix  $\tau$  and the point estimator  $\hat{\beta}$  are unchanged. Because the donor probabilities are differentiable in  $\psi$ , the smooth PMM block has an ordinary missing value score  $S_{\text{PMM},i}^{(p)}$ , and the standard observed score linearization for the predictive mean working model gives the nuisance influence contribution  $d_i^{(p)}$ . Stacking this block with the parametric blocks gives the same type of  $S_{\text{mis}}$  and  $d$  vectors as in Section A.1. The cross moment  $\kappa = E\{U S_{\text{mis}}^\top\}$  is computed from the rowwise completed record analysis contribution, not from the donor source score:  $\kappa$  measures the per record covariance between the analysis score and the imputation score, and is not affected by cross record correlation induced by donor sharing. The copied donor feature changes the independent first order summands for the direct analysis score covariance. When several recipients copy from the same observed donor, their analysis score contributions share the same stochastic source, so the independent summands are the donor source sums  $\tilde{U}_i$ . This replaces  $\Omega$  by  $\Omega_{\text{src}}$ . The same substitution applies to the direct analysis score side of the cross term, giving  $C_{\text{src}}$  in place of  $C$ , while the nuisance side remains  $\bar{d}_i$ . Under the stated moment and Lindeberg conditions, the empirical moments for  $\Omega_{\text{src}}$ ,  $\kappa$ ,  $\alpha$ , and  $C_{\text{src}}$  converge to their population analogues. The central limit theorem applied to the first order expansion and Slutsky's theorem then yield the stated sandwich limit and consistency of the sample analogue estimator. The Lindeberg condition rules out configurations in which a

single observed donor supplies values to a diverging fraction of recipients, causing one row of  $\tilde{U}^{(p)}$  to dominate the meat. In typical applications where the donor pool is large relative to the number of recipients from any one donor, this condition holds automatically.

## B Additional Simulation Results

Table 5 summarizes the reported  $n = 2000$  PMM donor-width sensitivity cells. Coverage of  $\beta^*$  remains close to nominal across the reported donor-width scales, whereas coverage of the data-generating coefficient decreases as the smooth donor law spreads probability over more distant donors. This pattern is consistent with a change in the PMM point-estimator target as the donor-width scale increases. Figure 3 shows the same  $n = 2000$  RW-PMM validation

| $k$ | Imputation<br>counts $m$ | Mean effective<br>donor count | Mean donor-pool<br>probability mass | Mean SE /<br>empirical SD | Coverage<br>$\beta^*$ | Coverage<br>$\beta_0$ |
|-----|--------------------------|-------------------------------|-------------------------------------|---------------------------|-----------------------|-----------------------|
| 5   | 5, 25, 50, 100           | 11.2                          | 0.59                                | 1.03–1.31                 | 0.954–0.979           | 0.934–0.968           |
| 10  | 5, 25, 50, 100           | 19.0                          | 0.66                                | 1.00–1.10                 | 0.946–0.956           | 0.923–0.944           |
| 20  | 5, 25, 50                | 32.5                          | 0.73                                | 0.97–1.02                 | 0.943–0.953           | 0.888–0.913           |
| 30  | 5, 25, 50                | 44.6                          | 0.77                                | 1.00–1.01                 | 0.946–0.952           | 0.862–0.886           |

Table 5: Reported  $n = 2000$  RW-PMM donor-width sensitivity summary. Entries are ranges across the imputation counts shown in the second column. Donor-pool probability mass is the probability the smooth law assigns to the  $k$  donors in the hard rule’s donor pool.

cells graphically across donor-width scales. Only cells with complete 2500-replicate results are plotted; [empty cells still running](#). Figure 4 shows the available RW-PMM validation cells graphically for donor-width scale  $k = 5$  across sample sizes. Only cells with complete 2500-replicate results are plotted. For the parametric validation analysis, the main text focuses on the original Giganti–Shepherd sample size  $n = 4000$ . Figure 5 shows the full parametric measurement-error grid for  $n \in \{2000, 4000, 6000, 8000\}$ , which is summarized in the main text only at  $n = 4000$ .

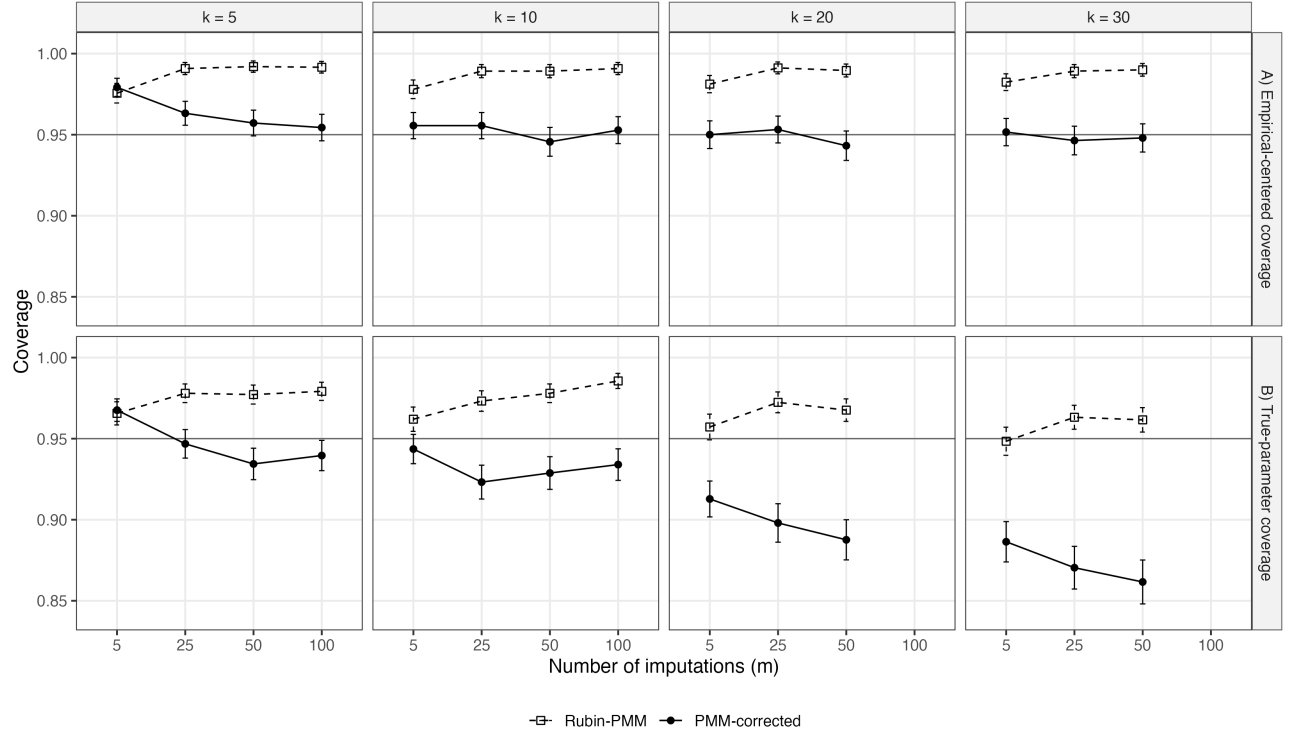


Figure 3: RW-PMM measurement-error validation results at  $n = 2000$ , shown across donor-width scales  $k$ . Rows distinguish coverage of  $\beta^*$  from coverage of  $\beta_0$ . Empty points mark simulation cells not included in the present result set.

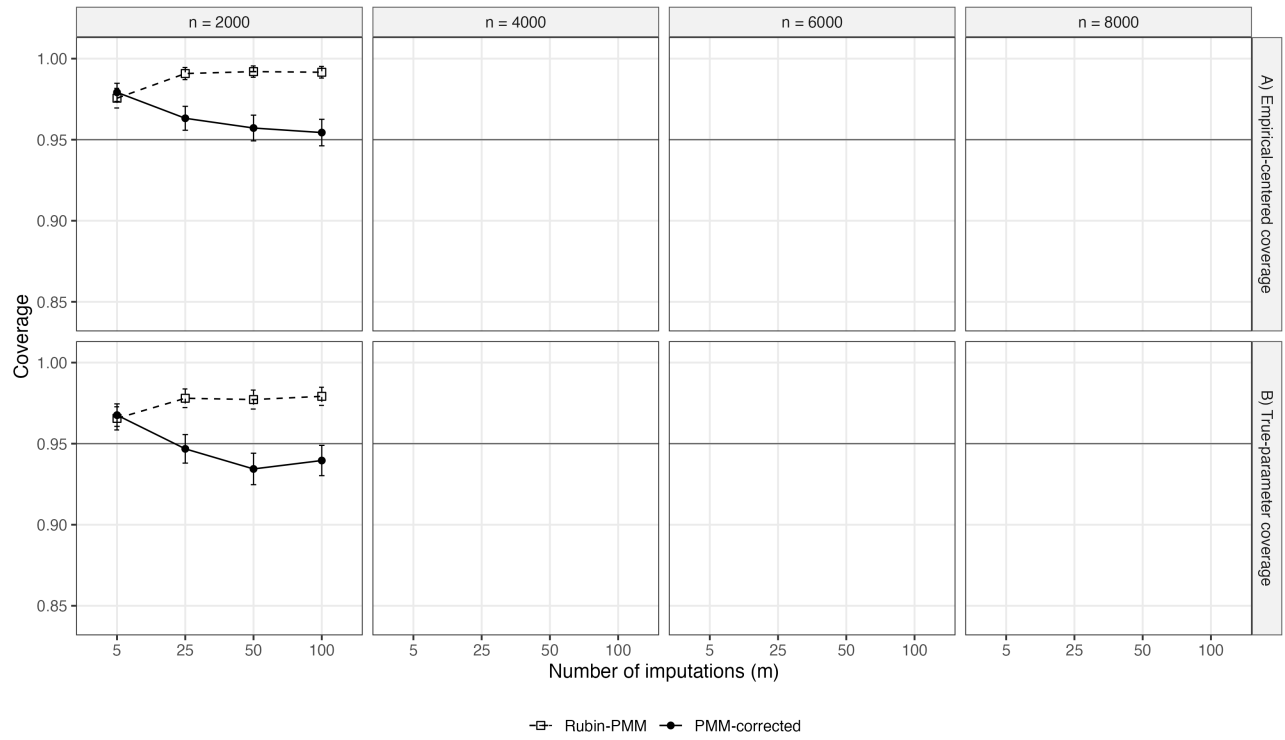


Figure 4: RW-PMM measurement-error validation results for donor-width scale  $k = 5$ . Columns show sample size  $n$ , and rows distinguish coverage of  $\beta^*$  from coverage of  $\beta_0$ . Empty panels mark simulation cells not included in the present result set.

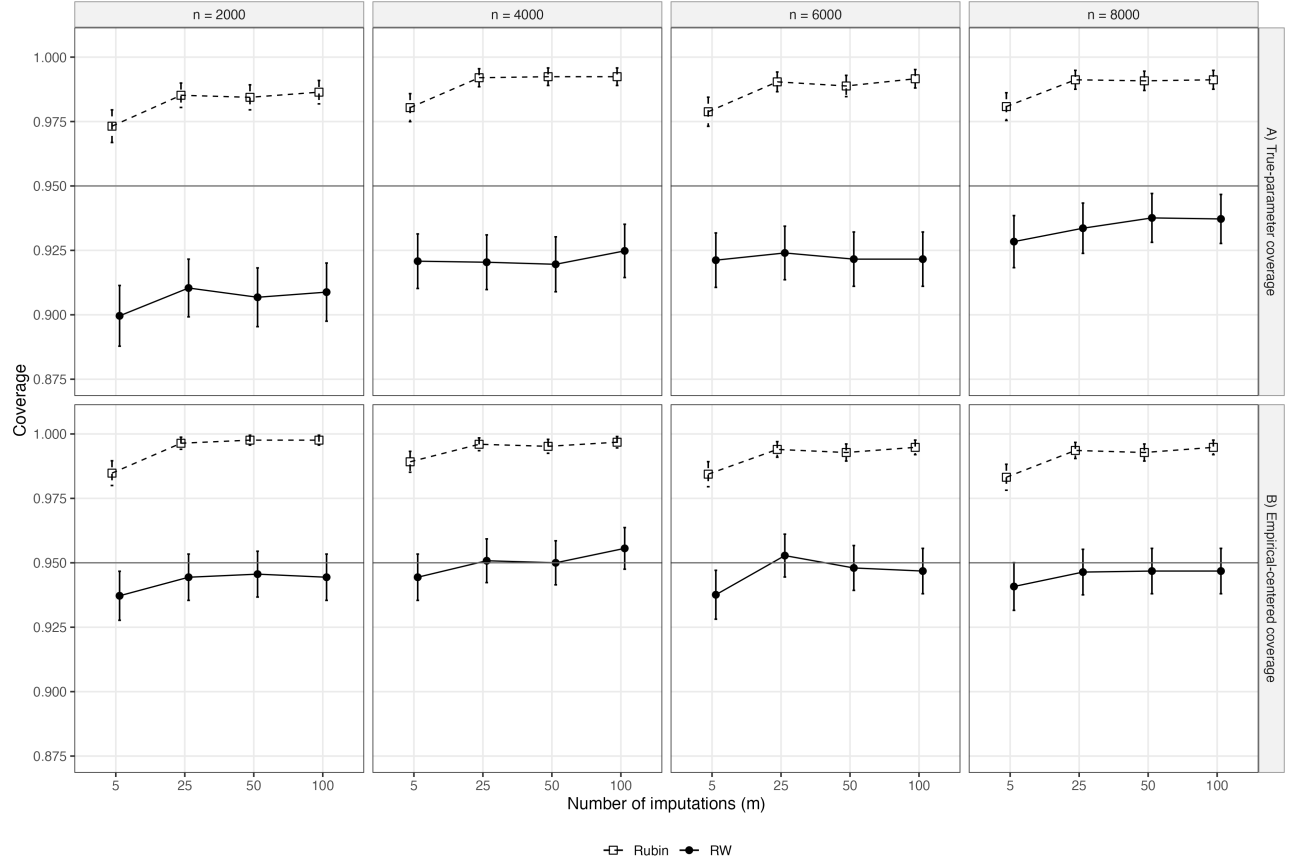


Figure 5: Full parametric MICE measurement-error simulation grid. Rows show coverage of  $\beta_0$  and coverage of  $\beta^*$  for the chained conditional MICE implementation. The main text focuses on  $n = 4000$ , matching the original Giganti–Shepherd sample size.



## References

- R. R. Andridge and R. J. A. Little. A review of hot deck imputation for survey non-response. *International Statistical Review*, 78(1):40–64, 2010. doi: 10.1111/j.1751-5823.2010.00103.x.
- J. W. Bartlett and R. A. Hughes. Bootstrap inference for multiple imputation under uncongeniality and misspecification. *Statistical Methods in Medical Research*, 29(12):3533–3546, 2020. doi: 10.1177/0962280220932189.
- J. W. Bartlett, S. R. Seaman, I. R. White, and J. R. Carpenter. Multiple imputation of co-variables by fully conditional specification: accommodating the substantive model. *Statistical Methods in Medical Research*, 24(4):462–487, 2015. doi: 10.1177/0962280214521348.
- M. J. Giganti and B. E. Shepherd. Multiple-imputation variance estimation in studies with missing or misclassified inclusion criteria. *American Journal of Epidemiology*, 189(12):1628–1632, 2020. doi: 10.1093/aje/kwaa153.
- M. J. Giganti, P. A. Shaw, G. Chen, S. S. Bebawy, M. M. Turner, T. R. Sterling, and B. E. Shepherd. Accounting for dependent errors in predictors and time-to-event outcomes using electronic health records, validation samples and multiple imputation. *The Annals of Applied Statistics*, 14(2):1045–1061, 2020. doi: 10.1214/20-AOAS1343.
- R. A. Hughes, J. A. C. Sterne, and K. Tilling. Comparison of imputation variance estimators. *Statistical Methods in Medical Research*, 25(6):2541–2557, 2016. doi: 10.1177/0962280214526216.
- J. K. Kim and W. A. Fuller. Fractional hot deck imputation. *Biometrika*, 91(3):559–578, 2004. doi: 10.1093/biomet/91.3.559.
- R. J. Little, J. R. Carpenter, and K. J. Lee. A comparison of three popular methods for handling missing data: Complete-case analysis, inverse probability weighting, and multiple imputation. *Sociological Methods & Research*, 53(3):1105–1135, 2024. doi: 10.1177/00491241221113873.
- X.-L. Meng. Multiple-imputation inferences with uncongenial sources of input. *Statistical Science*, 9(4):538–558, 1994. doi: 10.1214/ss/1177010269.
- J. N. K. Rao and J. Shao. Jackknife variance estimation with survey data under hot deck imputation. *Biometrika*, 79(4):811–822, 1992. doi: 10.1093/biomet/79.4.811.
- J. M. Robins and N. Wang. Inference for imputation estimators. *Biometrika*, 87(1):113–124, 2000. doi: 10.1093/biomet/87.1.113.
- D. B. Rubin. *Multiple Imputation for Nonresponse in Surveys*. John Wiley & Sons, New York, 1987.
- N. Schenker and J. M. G. Taylor. Partially parametric techniques for multiple imputation. *Computational Statistics & Data Analysis*, 22(4):425–446, 1996. doi: 10.1016/0167-9473(95)00057-7.
- M. Schomaker and C. Heumann. Bootstrap inference when using multiple imputation. *Statistics in Medicine*, 37(14):2252–2266, 2018. doi: 10.1002/sim.7654.

- J. Shen, K. A. Linn, A. S. Clark, R. Mamtani, and R. A. Hubbard. Missingness in eligibility criteria for target trial emulation in EHR with survival outcomes. *Statistics in Medicine*, 45, 2026. doi: 10.1002/sim.70500.
- C. J. Skinner. Jackknife variance estimation for multivariate statistics under hot-deck imputation from common donors. *Journal of Statistical Planning and Inference*, 102(1):149–167, 2002. doi: 10.1016/S0378-3758(01)00185-9.
- J. A. C. Sterne, I. R. White, J. B. Carlin, M. Spratt, P. Royston, M. G. Kenward, A. M. Wood, and J. R. Carpenter. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. *BMJ*, 338:b2393, 2009. doi: 10.1136/bmj.b2393.
- S. van Buuren and K. Groothuis-Oudshoorn. mice: Multivariate imputation by chained equations in r. *Journal of Statistical Software*, 45(3):1–67, 2011. doi: 10.18637/jss.v045.i03.
- N. Wang and J. M. Robins. Large-sample theory for parametric multiple imputation procedures. *Biometrika*, 85(4):935–948, 1998. doi: 10.1093/biomet/85.4.935.
- I. R. White, P. Royston, and A. M. Wood. Multiple imputation using chained equations: issues and guidance for practice. *Statistics in Medicine*, 30(4):377–399, 2011. doi: 10.1002/sim.4067.
- X. Xie and X.-L. Meng. Dissecting multiple imputation from a multi-phase inference perspective: what happens when God’s, imputer’s and analyst’s models are uncongenial? *Statistica Sinica*, 27(4):1485–1594, 2017. doi: 10.5705/ss.2014.067.
- S. Yang and J. K. Kim. Asymptotic theory and inference of predictive mean matching imputation using a superpopulation model framework. *Scandinavian Journal of Statistics*, 47(3): 839–861, 2020. doi: 10.1111/sjos.12429.